B.Comp. Dissertation

Optimizing WebNorm for Industrial Anomaly Detection: Addressing Explainability, Security, and Cost Challenges in Large Language Model Integration

By

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Abstract

Modern web applications, powered by web APIs (such as RESTful, SOAP, and WebSockets), are prime targets for attacks and unintended access, making anomaly detection crucial for their reliable operation. However, current log-based anomaly detection faces significant hurdles. Anomalous events can produce logs virtually indistinguishable from normal ones, often lacking vital context that might reside in databases. Moreover, noise within the log data can further impede the effectiveness of state-of-the-art log learning techniques, causing them to learn irrelevant patterns and build inadequate anomaly detection solutions.

In this work, we examine the industrial practicality of a newly proposed anomaly detection solution, WebNorm, whose performance rivals and even outperform state-of-the-art solutions. Then, we address its limitation by replacing the use of Large Language Models with traditional invariant generation tools in order to optimize WebNorm for use in industrial context. The optimised WebNorm is evaluated extensively on 25 types of web-tamper attacks on the benchmarks of Train-Ticket against the original WebNorm's baselines. The results showed that our optimization achieves high recall, indicating a promising direction for WebNorm future work.

Subject Descriptors:

- Intrusion/anomaly detection and malware mitigation
- Software security engineering

Keywords:

Anomaly Detection, Large Language Models, Web Application Security, Program Invariants, Software security engineering

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1. Introduction

In today's world, web applications are used in every sector, including private businesses and government. They facilitate essential services and interactions on a global scale. [1]As of 2023, over 64% of the global population regularly use the internet, with a substantial portion of the web traffic going to web applications. The vast reach of web application are demonstrated by the billions of visitors that popular platforms like [2]YouTube and Amazon attract every month. At such a scale, even minor bugs or security vulnerabilities can have major consequences in terms of cost to the organizations. Thus, robust web security measures are crucial to safeguard both organizations and their users.

1.1. The Role of Anomaly Detection

Among the various security measures that are available out there, anomaly detection plays a crucial role within the cybersecurity strategies of many organizations. The anomaly detection process works by pinpointing irregular patterns or deviations from typical operational trends within a system. These deviations can signal possible risks, such as cyber intrusions, fraudulent activities, system failures etc. Unlike conventional security techniques, which depend on predefined threats signature, anomaly detection can uncover new and previously unrecognized attacks, making it an invaluable asset in the fast-paced environment with constantly emerging threats.

For example, [3]Walmart uses a machine learning-based anomaly detection tool named AI Detect and Response (AIDR), which over a three-month period, identified 63% of major incidents and reduced the mean-time-to-detect[4] by over 7 minutes. The use of AIDR has enabled Walmart to respond quicker to potential threats, minimizing disruptions and reducing the cost associated with these incidents. This use case showed that such proactive threat identification provided by

[1]https://www.google.com/url?q=https://www.statista.com/topics/1145/internet-usage-worldwide/&sa=D&source=docs&ust=1730557175481613&usg=AOvVaw0xmJy5kdJO5zu6ThvY1fKV
[2] https://www.statista.com/statistics/1201880/most-visited-websites-worldwide/
[3]https://www.google.com/url?q=https://arxiv.org/abs/2404.16887&sa=D&source=docs&ust=1730557287644679&usg=AOvVaw0B2GMQ4tHfeb7MCeHK6N49
[4] https://www.techtarget.com/searchitoperations/definition/mean-time-to-detect-MTTD

anomaly detection systems is essential for minimizing the impact of cyber-attacks, demonstrating the importance of it in softwares.

2. Project Objectives

This project primarily aims to improve the practicality of WebNorm, a newly proposed anomaly detection framework, for use in industrial environments. Though WebNorm outperformed several state-of-the-art anomaly detection tools during initial testing, it currently encounters challenges when applied in real-world industrial settings. Industries such as finance and government have stringent non-functional requirements, and for WebNorm to be viable in these areas, it needs to address its current limitations that cause it to fail to meet these rigorous standards. In this project, we explore the current works of Webnorm, the challenges of implementing Webnorm effectively in an industrial context, as well as making improvements to Webnorm for industrial usage.

2.1. Industrial Setting Requirements

Industrial environments have specific requirements that are essential for the viability of any anomaly detection solution. This project focuses on 3 key requirements that have been identified as priorities by our partnering company.

1. **Security:** Amongst other security aspects, ensuring data privacy is paramount in industrial settings, especially when handling internal and/or regulated data. Anomaly detection tools must be able to comply with strict organizational data policies and/or government regulation, which includes the prevention of data leakage.
2. **Resource/Cost:** For any tool to be viable, it must operate efficiently within the organization's existing infrastructure and avoid excessive computational and/or financial overhead. This is particularly crucial in the industrial environment where scaling can exponentially increase costs. Therefore, an ideal detection framework requires minimal resources without compromising performance.
3. **Reliability:** In high-stakes environments, tools must be both consistent and interpretable. Correspondingly, reliability and transparency in anomaly detection’s results are essential for scaling and deployment without the risk of ineffectiveness.

2.2. Challenges with Large Language Model in WebNorm

Webnorm has great reliance on Large Language Models (LLMs) to achieve its anomaly detection capabilities. However, the integration of LLMs presents significant challenges in fulfilling the requirements set out in section 2.1:

1. **Explainability:** LLMs often function as “black boxes”, making it difficult to understand the logic behind their output. This lack of transparency can hinder trust in the tool and prevent users from confidently acting on its results, especially when regulatory compliance and operational decision-making is involved.
2. **Security Concerns**: If cloud-based LLM solutions are used, such as OpenAI, they may raise security issues particularly regarding handling of sensitive data. Giving data to cloud-based LLMs presents data privacy risks that are unacceptable for organizations bound by stringent data policy or government regulation. Ensuring data confidentiality and control over information processing is a top priority in such environments, making cloud-based LLMs a challenging choice.
3. **Cost and Resource Efficiency:** Deploying LLMs on a local infrastructure could mitigate some security concerns but introduces substantial costs in terms of infrastructure and maintenance. The opportunity cost of such infrastructure is especially high for Webnorm since LLM usage is limited to the pre-deployment phase, rendering LLM inactive in the operational phase. Such a level of costs to set up a local LLM may be prohibitive for organizations with limited resources.

Given these challenges, we aim to adapt WebNorm to meet the specific needs of industrial environments specifically by addressing the challenges posed by LLM integration.

3. Literature Review

This section reviews the different solutions related to our study. There are 2 main solutions that we will be discussing. The first is Webnorm, which is the newly proposed anomaly detection framework that we are improving on. The second is Daikon, which is an established invariant detector that can be used to identify likely program invariants.

3.1. Webnorm

To determine and implement meaningful improvements to WebNorm, we must begin by thoroughly examining its current framework and implementation methods. Although many anomaly detection methodology, especially highly performing ones like deep-learning and log-based techniques, have been developed and used on web applications, this project does not explore them. Instead, we focus on understanding WebNorm's underlying motivations and internal workings, as outlined in the following section.

3.1.1. Webnorm Motivation

WebNorm is an anomaly detection system that relies on the regular operating patterns, or "normality," of a web application to identify anomalies. By deriving invariants from typical behaviors and states, WebNorm establishes predefined rules for anomaly detection, providing a degree of explainability that traditional anomaly detection approaches often lack. Moreover, since WebNorm's invariants are created during the pre-deployment phase, the system is more resilient to gradual distribution shifts that commonly impact ML-based anomaly detection systems over time.

The primary focus of WebNorm is guarding against frontend tamper attacks[5], a category of cyber-attacks in which malicious actors manipulate client-side elements of a web application to alter its behavior. These attacks can be carried out in different ways, such as script injections, form modifications, or changes to API requests sent from the browser. These attacks are particularly challenging to detect because they occur on the client side where the users are in control, where traditional server-side detection methods, which focus on backend patterns, may

[5] https://owasp.org/www-community/attacks/Web_Parameter_Tampering

not capture the tampered interactions. By identifying differences from established invariants, WebNorm can detect anomalies indicative of tampering, even when these changes are subtle, and provide explanation using the violated invariants.

3.1.2. Webnorm Architecture and Internals

Webnorm revolves around 2 main underlying assumption:

1. A web application has a predefined set of normal states and behaviors, also known as its normality.
2. A web application has a pre-deployment testing phase.

In terms of normality, Webnorm focuses on learning 3 aspects of normality: data consistency, flow consistency, and common sense consistency. Data consistency refers to the equality of data values between 2 associated events. The author defined data consistency as follows:

Data Consistency:

Suppose each event e has its own corresponding API, denoted by e.API. Events are ordered by time using the precedence relation $<$, where $e_i < e_j$ indicates that e_i occurs before e_j.

We say there is a data consistency between two event types API_1 and API_2 if $\forall(e_i, e_j),\ e_i < e_j,\ e_i.API = API_1,\ e_j.API = API_2,$ $\exists p \in e_{i.\text{out}},\ q \in e_{j.\text{in}},$ such that $\text{value}(p) = \text{value}(q)$ where value(x) is runtime value of the parameter x.

To illustrate an example of data consistency, we consider a simple sequence of events in the backend that represents the workflow of purchasing items:

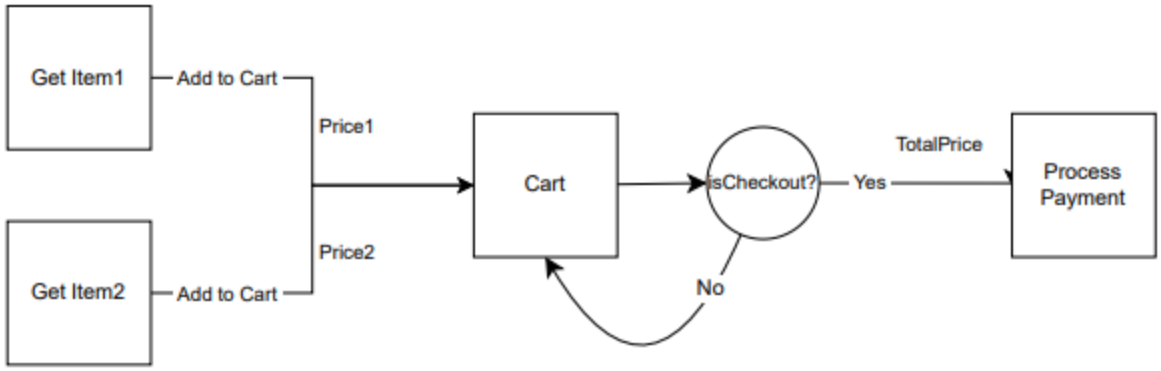

Figure 1: Item Purchase Event Flow

Data consistency for Price 1, Price 2 & Total Price are crucial. For instance, the TotalPrice that is received by the Process Payment event must be equivalent to the TotalPrice outputted by the Cart. Any discrepancy could indicate a system bug where the TotalPrice is not given by the Cart directly.

Flow consistency refers to the validity of the sequential order of events. It is defined as

Flow Consistency:

We say there is flow consistency between two event types API_1 and API_2 if $\forall \langle e_i, e_j \rangle,\ e_i < e_j,\ e_i.\text{API} = API_1,\ e_j.\text{API} = API_2,$ $\exists$cond as a condition defined on e_i.in so that if $\text{cond}(e_i.\text{in}) = \text{true}$, e_j needs to appear in the log sequence within k events after e_i appears.

With reference to figure 1, if condition "isCheckout" evaluated to true, then a Process Payment event should follow in the subsequent order of events. The absence of the Process Payment may indicate a system anomaly where payment is bypassed.

Common-sense consistency refers to the validity of the inputs to an event. It is defined exactly as:

Common-sense Consistency:

We say there is common-sense consistency on an event type API if $\forall e,\ e.\text{API} = API,\ \exists$cond as a condition defined on e.in so that $\text{cond}(e.\text{in})$ is always true.

A common-sense consistency in figure 1 could be “TotalPrice >= 0” since paying for a negative price constitutes an anomaly behavior.

These normalities provide a well-defined scope for detecting anomalies which is much more practical than guarding against novel attacks with undefined boundaries. To acquire these normalities, a pre-deployment testing phase is required for Webnorm to perform the following

1. **Program Analysis:** Program analysis uncovers normalities related to event relations through the construction of an Event Graph. For example: Can a “checkout” event be directly followed by a “order tracking” event without an intermediate “payment” event?
2. **Log Collection:** Logs represent partial snapshots of the web application at a given point in time. They are used to supply runtime parameter values to static event definitions. For example: What is the range of values that is valid for the total price of a cart?
3. **Large-Language Model Analysis:** LLM is used to learn the normalities from the results of the logs and program analysis. To represent these normalities, Webnorm utilizes LLM to construct first-order logic on logs, which directly corresponds to rules that are used in rule-based anomaly detection systems.

These 3 components are architected as below:

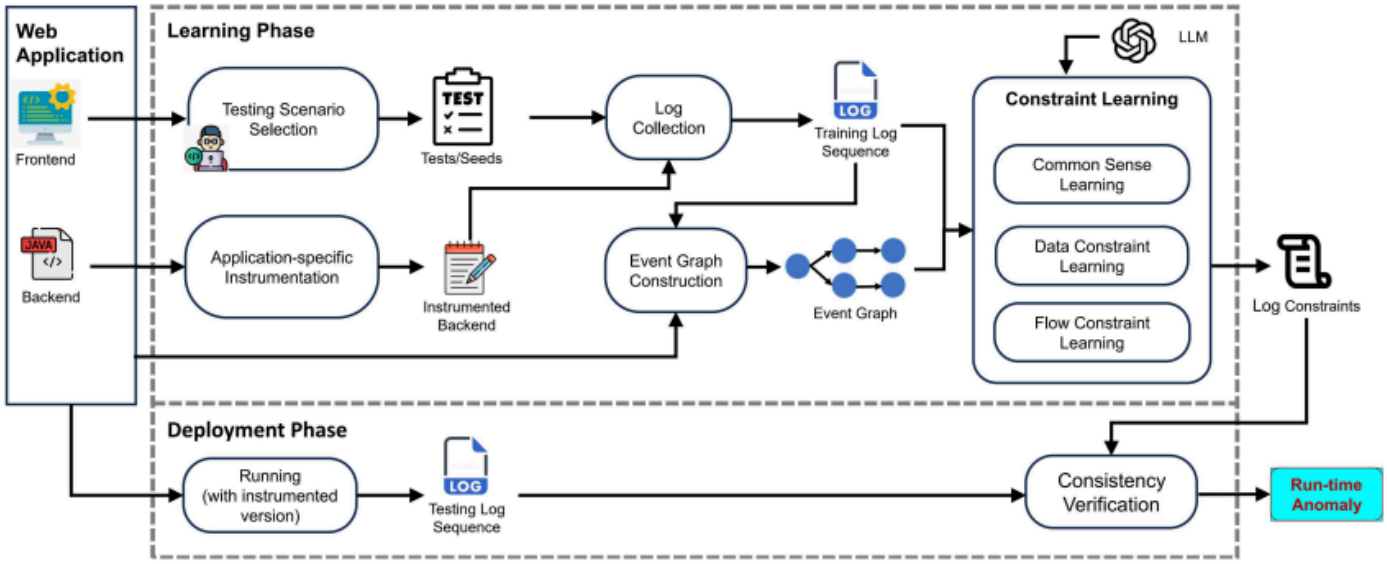

Figure 2: Webnorm Architecture[6]

[6] https://jasonbourne1998.github.io/data/ASE24.pdf

This architecture is important because it is used to link between events from the program analysis and the raw logs collected. Raw logs provide runtime parameters and values which can be mapped to their corresponding events, allowing Webnorm to identify related logs amongst a large volume of logs, and learn the relationship between them.

3.1.3. Webnorm Invariants Generation

Webnorm relies on LLM to generate invariants, introducing the challenges discussed in section 2.2. To address these challenges, we will analyze in detail to understand the invariant generation core functionality and the necessary requirements for viable alternatives.

Given a group of events, consisting of extracted logs and event relations, the LLM is prompted to infer invariants in the form of first-order logic, and output a Python function that checks the invariant. The output invariants are made more accurate and correct by using different prompt engineering techniques.

 # Identity
You are a software engineer that is extremely good at understanding business logic and user requirements.
You are responsible for writing functions to validate the correctness and usefulness of input data.
At each turn, you should first provide your step-by-step thinking for solving the task. Your thought process should be enclosed using "<thought>" tag.
After that, you should use write a function in a block of Python code to solve the task.

Prompt 1: System prompt to generate common-sense invariants

To increase accuracy, a chain-of-thought approach is used by prompting the LLM to provide step-by-step output. This guides the model through each logical step, ensuring a more reliable and coherent output. With reference to Prompt 1, LLM are guided to output intermediate results, enclosed using “<thought>” tag. These intermediate results are extracted out of the responses and discarded as they serve no purpose to Webnorm in the deployment phase.

 # Identity

You are a software engineer that is extremely good at modelling entity relationships in databases. You are responsible for checking how two different classes are related to each other.

At each turn, you need to do THREE things:

1. You SHOULD first provide your step-by-step thinking for solving the task. Your thought process should be enclosed using "<thought>" tag.

2. You SHOULD construct as many of the most important first-order logic constraints and output it in general format. Examples:

- $\forall x$ *(isDog(x)* $\rightarrow$ *hasFourLegs(x)*
- $\exists x$ *(isCat(x)* $\wedge$ *isBlack(x))*
- $\forall x$ *(isPerson(x)* $\rightarrow$ $\exists y$ *(isDog(y)* $\wedge$ *owns(x, y)))*
- $\forall x$ $\forall y$ *((isParent(x, y)* $\wedge$ *isMale(x))* $\rightarrow$ *isFather(x, y))*
- $\exists x$ *(isHuman(x)* $\wedge$ *loves(x, Mary)),*
- $\forall x$ *(isStudent(x)* $\wedge$ *studiesHard(x)* $\rightarrow$ *getsGoodGrades(x))*
- $\forall x$ *(isAnimal(x)* $\rightarrow$ *(*$\exists y$ *(isFood(y)* $\wedge$ *eats(x, y))))*

)

3. You SHOULD write a function in a block of Python code to solve the task based on the constraints.

Prompt 2: System prompt to generate input invariants

Additionally, WebNorm incorporates in-context learning by embedding similar examples directly into the prompt. In Prompt 2, examples of data dependency representations are given with specific enclosing tags so that they can be easily extracted out using regex that identifies the surrounding enclosing tags.

Your code failed {fails} test cases. There should be AT LEAST ONE MATCH. Please try to correct the wrong condition or RELAX THE CONSTRAINTS to match instances from [A] to [B], and try again.
- You SHOULD return True if all checks passed, otherwise, raise Error on the specific violation with a detailed message (NO NOT use any return (E.g return False / return instance_A['authorization'] == instance_B['authorization']) here !! Directly Raise error).

{reasons}

Prompt 3: User prompt to provide feedback for invalid input invariants

A few-shot learning strategy is also employed, involving a prompt-refining step that iteratively enhances the prompt to yield correct results as exemplified in prompt 3. Lastly, a training-and-validation approach minimizes hallucinations and improves output quality by assessing generated invariants against a validation set, refining both prompts and models to ensure robust invariant generation.

To remove the reliance on LLM, we look to traditional invariant detection tools, one of which is Daikon.

3.2. Daikon

Daikon is a dynamic analysis tool designed to detect likely invariants in programs by observing their executions. The full process of detecting invariants involves three primary steps:

1. **Instrumentation**: The target program is instrumented to insert probes that record variable values during execution. This requires access to the underlying program code and supports many languages.
2. **Execution**: The instrumented program is run to generate execution traces dynamically. This is a crucial step that determines the accuracy of the resulting invariant. Thus, the program should be executed to cover a complete and quality test suite, which can be crafted through existing test suite crafting techniques[7].
3. **Inference**: Daikon analyzes the execution traces to infer invariants by checking candidate invariant patterns against the recorded traces. An invariant is deemed accurate if the number of failed checks does not exceed a threshold, sufficiently determined by Daikon.Thus, invariants are termed “likely” as they are only statistically justifiable by the given set of traces from the execution phase.

Daikon can infer a variety of invariant types, including constants, range constraints (variables within specific bounds), linear relationships relating multiple variables and conditional invariants, all of which are useful in WebNorm’s use case.

3.2.1. Daikon Architecture

Daikon's architecture closely mimics its invariant detection process and comprises several components working independently:

1. **Front-end Instrumentation Tools**: Language-specific tools that instrument the target program. For Java programs, Daikon uses Chicory.
2. **Execution Trace Generation**: The instrumented program produces execution traces, recorded in Daikon's dtrace file format.
3. **Invariant Detection Engine**: Daikon's core component that reads the execution traces and infers invariants.

[7] https://homes.cs.washington.edu/~mernst/pubs/daikon-tool-scp2007.pdf (page 42)

The modular design allows Daikon to support multiple programming languages by implementing appropriate front-end tools. The invariant detection engine remains language-agnostic, operating solely on the execution traces.

3.2.2 Trace Generation

The problem arises as Daikon's trace generation requires an instrumentation on the program code that is carried out in the 1st phase, which may not be feasible in industrial settings as it possesses risk and privacy issues. To continue leveraging Daikon's invariant detection engine, we look into trace generation without access to underlying implementation. In particular, we will be looking into the use of HTTP traffic to generate traces. Before which, understanding the trace generation process is crucial for extending Daikon's capabilities to new data sources, in this case, HTTP traffic. Since Webnorm is evaluated using Java applications (Train-ticket[8]), we will be analyzing Daikon's front-end tool for Java program, Chicory. Note that the theoretical concept of the different front-end tools are similar.

Chicory instruments Java bytecode to insert probes that capture variable values at runtime. The instrumentation focuses on 2 aspects. Firstly, method entries/exits are crucial points in the program, and parameter values upon method entry and return values upon method exit are captured. Secondly, variables that have been identified to be of interest, either through specification by the user or default settings, will be captured as well. These variables can include class fields, local variables etc.

The instrumentation also involves inserting calls to Daikon's runtime library for data recording. Once instrumented, the Java program is executed normally and the inserted probes collect variable values throughout the program and write them to a dtrace file, following Daikon's trace file format.

[8] https://github.com/FudanSELab/train-ticket

4. Work Done

Based on the 2 literature reviews, the performing invariant detection engine of Daikon is an attractive option to be used for invariant generation. As such, we have decided to incorporate Daikon into Webnorm, replacing the use of LLM in its invariant generation process. The project is split into 3 core phases. The first phase involves the actual implementation of invariant generation using Daikon. The second phase involves the generation of various datasets in preparation for the next phase. The last phase is the evaluation of our approach using the generated datasets.

4.1. Invariants

In this first phase of the project, we will dive deep into our proposed solution to infer invariants from HTTP traffic data and transform these invariants into runtime checking tools.

4.1.1. Extending DTrace generation to HTTP Traffic Data

To enable invariant detection on HTTP traffic using Daikon, we propose modeling HTTP logs as program points in the DTrace format. This approach allows us to treat structured HTTP data similarly to runtime values in conventional programs, enabling analysis via Daikon's inference engine.

A DTrace file consists of three primary components: **(A) Declarations**, **(B) Program Point Definitions**, and **(C) Data Trace Records**. Each component plays a specific role in defining the structure of the data and enabling Daikon to generate meaningful invariants. We describe our modeling strategy for each component below.

A. Declaration

The declaration section contains metadata about the trace file. Our approach selectively includes relevant fields, omitting those that are either redundant or incompatible with HTTP traffic modeling as described in the following:

```
// Metadata
decl-version 2.0
```

Figure 3: screenshot of the declaration section of a DTrace file

- *Declaration version*

  We adopt the latest version of the DTrace format: “decl-version 2.0”.

- *Input-language*

  This field specifies the source language. As it is arbitrary and unused by Daikon, we exclude it from our trace files.

- *Variable comparability*

  This field indicates comparability relationships between variables. Since HTTP fields lack semantic links in log format, we exclude this to avoid misleading comparisons. Future study may involve ways to infer such semantic relationships between fields and utilise this field.

- *ListImplementors declaration*

  While this field is designed to support user-defined classes that implement the java.util.List interface, it is unsuitable for complex structures like JSON in HTTP logs. Our experiments reveal that Daikon fails to process such structures effectively, even when properly declared. As a result, this field is omitted, and complex nested fields are handled through flattening which will be discussed later.

B. Program point declarations

Each program point defines a unique context (akin to a function or event) and the variables associated with it. HTTP logs are treated as such contexts, with each API call corresponding to a program point.

- *Program point name*

  The name uniquely identifies a program point. Since Daikon does not analyze the name semantically, we standardize on the placeholder: “Main.main():::Point”. To facilitate focused invariant inference, each DTrace file will contain a single program point. This separation aids debugging and isolates invariants to specific HTTP endpoints.

- *Program point type*

  For data not originating from a programming language (such as HTTP logs), Daikon mandates the “point” type. Unlike the other types, this type confines invariant detection to within a program point, without leveraging Daikon’s inter-point dataflow analysis. Hence, to capture invariants across multiple HTTP entries, related logs must be grouped under the same program point.

- *Variable Declaration*

  Each variable within a program point is described by a block containing several attributes:

  - *Variabel Name*

    An arbitrary identifier.

  - *Variable Kind*

    Amongst the various kinds supported by Daikon, only 2 kinds of variables are relevant to HTTP logs: “array” and “variable”.

  - *Enclosing Variable*

    Required for array kinds; specifies the name of the parent scalar variable.

- *Array Dimension*

  This field has default value 0. However, for “array”-kind variables, an integer value should be specified to indicate the number of dimensions that the array has.

- *Declaration Type*

  An arbitrary string that can be user-defined. We will follow Java’s naming for standard types (e.g. int, boolean, java.lang.String, etc).

- *Representation Type*

  Unlike declaration type, representation type is the actual type that will be used in the invariant generation process. Different types are handled differently, resulting in different invariants. The representation type must be one of boolean, int, hashcode, double, or java.lang.String; or an array of one of those (indicated by a [..] suffix). For the enclosing variable of arrays, we shall use the “hashcode” type. This indicates to daikon that these variables, and their values, are considered uninteresting for invariants detection.

To give an example of program point declaration, let's consider a HTTP log of an API (named “OrderQuery”) that query for the list of orders belonging to an user shown in figure 4.


```
1  {
2      "status":1,
3      "msg":"Query Orders For Refresh Success",
4      "orders":[
5          {
6              "id":"a100",
7              "price":100.00
8          },
9          {
10             "id":"b101",
11             "price":200.0
12         }
13     ]
14 }
```


Figure 4: HTTP Log instance for OrderQuery API

The program point declaration that corresponds to this HTTP log is shown in figure 5.

```
// Declarations
ppt Main.main():::Point
    ppt-type exit
    variable status
        var-kind variable
        dec-type int
        rep-type int
    variable msg
        var-kind variable
        dec-type java.lang.String
        rep-type java.lang.String
    variable orders
        var-kind variable
        dec-type java.lang.Object[]
        rep-type hashcode
    variable orders[..].id
        var-kind array
        enclosing-var orders
        array 1
        dec-type java.lang.String[]
        rep-type java.lang.String[]
    variable orders[..].price
        var-kind array
        enclosing-var orders
        array 1
        dec-type double[]
        rep-type double[]
```

Figure 5: Program Point Declaration for OrderQuery API

For the “orders” field in the HTTP log, it is an array of json. However, as lists of user-defined classes are problematic due to the limitation with ListImplementors as discussed earlier, we will handle such json by flattening the array objects and separating the fields into different variables. In particular, the variable “orders[..].id” is an array of String consisting of all order IDs, the variable “orders[..].price” is an array of doubles consisting of all order prices, and the same index in these arrays correspond to the same order object. Both arrays are enclosed by the scalar variable orders, which is declared with rep-type hashcode to indicate its role as a structural placeholder.

C. Data trace records

Data trace records provide the actual runtime values corresponding to a specific program point. Each record includes:

- *Program Point Name*

  The program point that this data trace record corresponds to. With reference to our program point name discussion, we will use "Main.main():::Point" only.

- *Variables*

  For each declared variable, the following fields must be provided in the same order as declared:

  - Variable Name

    The corresponding matching variable of the program point.

  - Variable Value

    The integer/boolean/string/array/etc value representation.

  - Modification Bits

    This value must be either 0/1/2. With reference to Daikon's documentation[9], it is safe to always set it to 1, which will be the case in all our DTrace.

[9] https://plse.cs.washington.edu/daikon/download/doc/developer.pdf (page 38)

```
// Trace Records
Main.main():::Point
status
1
1
msg
"Query Orders For Refresh Success"
1
orders
12345
1
orders[..].id
["a100" "b101"]
1
orders[..].price
[100.00 200.00]
1
```

Figure 6: Data Trace Records for OrderQuery API

Continuing from the previous OrderQuery example, the corresponding trace record is shown in Figure 6. It includes values for orders[..].id, orders[..].price, and any other declared fields. The full example DTrace is attached as Appendix A.

4.1.2. Automatic generation of DTrace using HTTP Traffic Log

Given a large dataset of HTTP logs, we want to be able to generate DTrace files automatically based on the algorithm discussed in section 4.1.1, before passing it to Daikon for invariants generation.

However, a major challenge arises from the lack of a standardized format for HTTP requests and responses. Each developer may define their own schema and semantics, leading to variability in fields, data types, and even the presence or absence of optional fields.

To address these issues, we introduce a schema inferring component to reverse-engineer the schemas involved. In this section, we will elaborate on an implementation[10] of this component that targets RESTful APIs due to its popularity in web applications and for use in our evaluation.

First, with reference to Appendix B, we defined several data types to capture the semantics of an arbitrary data field. These data types are essentially finite as they are each mapped to a "Representation Type" supported by daikon discussed in section 4.1.1. This allows a variable of such data types to be converted into a variable declaration in DTrace directly.

Next, logs are parsed and grouped into their respective API endpoints. Additional metadata such as sequence number and parsed timestamps are appended, allowing each entry to be uniquely identified and ordered accordingly later on. For each field in the log, the schema inducer tracks and collects different properties. For primitive types like strings, integers, floats, booleans, and bytes, it collects statistics such as length, value range, duplicate values, nullability etc. For composite types like arrays and objects, the inducer recursively processes their elements or fields.

Based on the observed properties, the inducer assigns the most suitable data type to each field, creating a specific schema. This method of inferring data types through properties accounts for variability; reverse engineering a schema does not guarantee 100% accuracy, thus, to avoid conflicting schema given new logs in the future, we dynamically analyse a large dataset (to get as close as possible to a complete dataset) to infer their semantics and account for any inconsistencies before defining it.

Also, the inducer may allow for the conversion of values into other types based on their observed properties, ensuring that invariants are meaningful. A commonly seen example is the "price" field, which are often stored as strings to avoid precision issues. Converting price string values into double values allows arithmetic-related invariants to be inferred.

Recall that user-defined object types are not supported in daikon and these complex objects should be flattened in DTrace. The inducer next step involves transforming nested object types

[10] https://github.com/ngyida/webnorm-lite/tree/master/webnorm_gpt/schema_induction

(if any) into a flattened representation by recursively extracting out fields in an array/object data types until arriving at a standalone primitive type.

	status	msg	orders	orders[..].id	orders[..].price
Log 1	1	Query Orders For Refresh Success	12345	[“a100”, “b101”]	[100.00, 200.00]
Log 2	0	Query Orders For Refresh Failed	None	None	None

Table 1: Schema and log instances for example OrderQuery API

A schema is outputted and represented by a Table. The columns represent the flattened fields of the logs, all of which have primitive type. Each row represents an instance of HTTP logs. With reference to Table 1, Log 1 corresponds to the HTTP Log in figure 4. Noting the similarity between Table 1 and Figure 5, a table can be easily translated into DTrace format. Thus, given a set of logs, we are able to automatically infer API schemas and create schema tables, then convert these tables into DTrace files.

4.1.3. Identifying Related Events in Logs

Due to the limitation discussed in section 4.1.1 whereby we are limited to “point” as our program point type, Daikon will only infer invariants based on the data contained within individual program points. To derive invariants that capture relationships across multiple HTTP events, it is essential to group related API calls into a single program point. This grouping enables us to identify interactions between API events that may span multiple requests, which is particularly important and common in web environments.

Given a large dataset of HTTP logs, we must identify related APIs to subsequently form composite program points that combine log instances of related events, allowing Daikon to infer more meaningful invariants that capture cross-event behaviors.

Modeling User Behavior with Non-Deterministic Automata

Web frontend API requests often exhibit structured patterns driven by user behavior. These patterns manifest as sequences of actions—for instance, navigation through a website, execution of common workflows, or repeated sequences of requests during a typical user session. Such behavior is inherently stochastic and exhibits variations that are well-suited for modeling with non-deterministic automata.

Hidden Markov Models (HMMs) share key mechanisms with non-deterministic automata, including transition between states, emission probabilities of a given state and initial state distribution. Thus, HMMs provide a possible fit for our use case. In particular, HMMs could achieve the following in our context:

- *State Estimation:*
  Hidden states can be interpreted as phases in a user's interaction with the web application. For example, a user might transition from a login phase to a browsing phase and then to a checkout phase.

- *Sequence Modeling:*
  By learning the probabilities of transitioning between states and the likelihood of each API request being produced by a given state, the HMM encapsulates the sequential dependencies and probabilistic structure of user behavior.

- *Handling Variability:*
  Due to the non-deterministic nature of HMMs, they can accommodate the inherent variability in user behavior. Even if two users follow similar workflows, slight differences in the sequence or timing of API calls are naturally modeled by the probabilistic transitions of the HMM.

Learning Event Relations using HMMs

In our implementation[11], the first step to training our HMM is to transform raw HTTP logs into a structured sequence by mapping each API into a unique numerical representation. The HMM is configured with a number of hidden states proportional to the number of API. Afterwhich, the HMM is trained on the HTTP logs dataset to learn the state transition probabilities and the emission probabilities for each API in each state.This training process leverages iterative optimization to adjust the model parameters, ensuring that the inferred hidden states best explain the observed sequences of API calls.

After training the HMM, we perform a backward propagation to iteratively propagate the influence of states backward through the model, before arriving at the influence value of each state on other states. The influence value is directly used to determine the strength of association between 2 API.

4.1.4. Joining Related Logs

Given a pair of related APIs taken from the HMM results, we achieve the grouping of this pair of APIs by joining their respective API tables based on temporal proximity and Context Matching.

1. *Temporal Proximity:*
   For every log entry in the first table, its timestamp is compared against those in the second table. The algorithm searches for the most recent log entry in the second table that occurs before the first table log's timestamp. To ensure relevance, a time threshold is applied (e.g., within 10 minutes). This constraint helps to avoid linking events that are too temporally distant.

2. *Context Matching:*
   In addition to time, the join condition also requires a match in a key shared attribute to verify that the two log entries belong to the same user session or context, e.g authorization header.

[11] https://github.com/ngyida/webnorm-lite/tree/master/webnorm_gpt/hmm_deduction

The output of grouping these related events are newly combined tables, with columns from both API and rows representing instances of paired logs.

```
1  {
2      "is_cancelled": true,
3      "order_id":  "a100"
4  }
```

Figure 7: HTTP Log instance for CancelOrder API

Let's consider another HTTP log of a different API (named “CancelOrder”) that cancels an order as shown in figure 7.

	status	**msg**	**orders**	**orders[..].id**	**orders[..]. price**	**Is_ cancelled**	**order_id**
Log 1	1	Query Orders For Refresh Success	12345	[“a100”, “b101”]	[100.00, 200.00]	true	“a100”
Log 2	0	Query Orders For Refresh Failed	None	None	None	None	None

Table 2: Schema and log instances for OrderQuery-CancelOrder API pair

Since it is common for the frontend to first load the list of orders using OrderQuery API, then subsequently let the user select an order for cancellation using the CancelOrder API, the trained HMM could identify them to be closely related and they could be grouped to form Table 2.

4.1.5. Invariants Generation

At this point, we have 2 types of schema tables - one generated directly from individual APIs, the other from combining 2 related tables. We convert these tables into DTrace files automatically as described in section 4.1.2. Then, Daikon's invariant detection engine analyzes the DTrace file and outputs a set of invariants. Among the various output formats supported by Daikon, we choose the Java format which produces actual Java code snippets that represent the invariants. Such programmatic representation allows us to further check invariants against real-time logs for monitoring as discussed in the next section.

For example, running the DTrace file from section 4.1.1 (Appendix A) will output the invariants shown in figure 8.

```
PS C:[redacted]\daikon-5.8.20> java -cp dai
kon.jar daikon.Daikon --format Java report.dtrace
Daikon version 5.8.20, released May 14, 2024; http://plse.cs.washing
ton.edu/daikon.
Processing trace data; reading 1 dtrace file:
[2025-04-07T00:03:43.358270]: Finished reading report.dtrace
===================================================================
=======
Main.main():::Point
status == 1
msg.equals("Query Orders For Refresh Success")
orders != null
daikon.Quant.pairwiseEqual(orders, new String[] { "a100", "b101" })
daikon.Quant.subsetOf(orders, new String[] { "a100", "b101" })
daikon.Quant.pairwiseEqual(orders, new double[] { 100.0, 200.0 })
daikon.Quant.subsetOf(orders, new double[] { 100.0, 200.0 })
Exiting Daikon.
```

Figure 8: Daikon Output in Java format

Note that the invariants utilise a Java package "Quant"[12] from Daikon.

[12] https://plse.cs.washington.edu/daikon/download/api/daikon/Quant.html

4.1.6. Converting Invariants into Runtime Checkers

Given that we have a set of invariants, written in snippets of incomplete Java code, we have to apply them in real time, supporting dynamic analysis and anomaly detection in our system. To do so, we use a template-based code generation solution to create runtime checkers i.e. a software program that takes in a log input, checks the input against the set of invariants, and outputs the violations if any.

We define a series of Java code templates that encapsulate each invariant as a method that can be executed at runtime. These checker classes are designed to accept log instances and determine if the invariant holds true.

The actual Java Template are given illustrated below:

1. **Template for a Single Invariant Checker**

```java
package daikon_inv_gen;

@FunctionalInterface
public interface InvChecker {
    public boolean check(InvVariables vars);
}
```

```
INV_SUB_CLASS_TEMPLATE = """
public static class InvChecker{idx} implements daikon_inv_gen.InvChecker {{
    @Override
    public boolean check(daikon_inv_gen.InvVariables webnorm_invariant_gen_vars) {{
        // {expr_original}
        return {expr};
    }}
}}
"""
```

2. **Template for Multi-Invariants Checker**

```
package daikon_inv_gen;

public interface InvCheckerAll {
    public InvChecker[] getInvCheckers();
}
```

```
INV_CLASS_TEMPLATE = """
public class InvCheckers implements daikon_inv_gen.InvCheckerAll {{
    {sub_classes}

    @Override
    public daikon_inv_gen.InvChecker[] getInvCheckers() {{
        return new daikon_inv_gen.InvChecker[] {{
            {inv_checkers}
        }};
    }}
}}
"""
```

Using one of the invariant from figure 8 - *msg.equals("Query Orders For Refresh Success")* - the InvCheckers and InvChecker class generated is illustrated below:

```java
package daikon_inv_gen;


public class InvCheckers implements daikon_inv_gen.InvCheckerAll {

    public static class InvChecker0 implements daikon_inv_gen.InvChecker {
        @Override
        public boolean check(daikon_inv_gen.InvVariables webnorm_invariant_gen_vars) {
            // msg.equals("Query Orders For Refresh Success")
            return ((webnorm_invariant_gen_vars
            .getStr(idx:1))
            .equals ( anObject:"Query Orders For Refresh Success" ));
        }
    }

    // other inv checker implementation can be added here

    @Override
    public daikon_inv_gen.InvChecker[] getInvCheckers() {
        return new daikon_inv_gen.InvChecker[] {
            new InvChecker0()
            // other inv checker instances can be added here
        };
    }
}
```

With a given InvCheckers object, we can instantiate a runtime checker that reads in HTTP logs that have been schematized using the schema inducer. The runtime checker checks each log instance against every single invariant checker and records the results. The implementation [13]of the core checking logic is attached in the Appendix C.

[13] https://github.com/ngyida/webnorm-lite/tree/master/spring-boot-analyze/daikon_inv_gen

4.2. Dataset

Before we are able to evaluate the effectiveness of our solution, we need to generate real HTTP traffic logs from a web application. To align with the baseline used in WebNorm, we collected our datasets from the same web application, Train-Ticket, which is widely used in the web development community for benchmarking.

In our approach, Daikon infers invariants from input logs. This approach requires the input logs to be complete and cover all use cases in order for meaningful and accurate invariants to be found. To ensure a comprehensive dataset, we constructed scripts[14] to automate the generation of HTTP traffic logs.

4.2.1 Training Dataset

Training logs are used by Daikon to infer invariants. For that, we identified several normal operations and script each operation. We then construct a finite state machine to chain these operations.

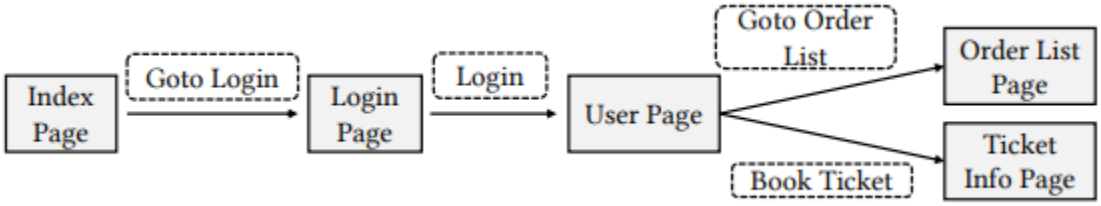

Figure 9: : A subset of the finite state machine for the TrainTicket dataset.

Figure 9 illustrates a portion of the Train-Ticket system's finite state machine. Solid-lined rectangles denote states, and dashed-lined rectangles on arrows represent actions. The example highlights four actions: Goto Login, Login, Goto Order List, and Book Ticket. Some actions, like Goto Login and Goto Order List, are simple redirects or clicks. Others, such as Login (entering credentials) and Book Ticket (filling details), involve more complex interactions with the web application. This finite state machine defines the specific sequence in which these actions occur.

[14] https://github.com/ngyida/webnorm-lite/tree/master/dataset/train_ticket

To implement user actions, we use the Selenium WebDriver[15] to interact with the web application and execute the scripts. During execution, the scripts randomly select an action based on the current state and execute it.

4.2.2 Test Dataset

Test logs are checked against the various invariants inferred from the training logs. To evaluate the effectiveness of our approach, test logs consist of various attacks which should ideally be identified through the violation of invariants. Thus, for each functionalities of Train-Ticket, we identified possible attacks and wrote scripts to perform these attacks using Selenium WebDriver to interact with the web application and mitmproxy[16] to intercept and modify the requests. The details of the 25 types of attack can be found in the Appendix D. Here, we will focus on 2 selected types of attack as an illustration.

Attack A

This attack involves the API to place consignment orders(/api/v1/consignservice/consigns). In this API, an important field is the“orderId” field in the request payload which refers to the order ID that the consignment will be added to. In its implementation[17], there is a lack of order ID field validation which can easily lead to a frontend tamper attack. In this attack, we intercepts and modifies the API request to tamper with the “orderId” in the request payload, replacing it with another user's Order ID, gaining unauthorized access to place consignment orders for orders that do not belong to the attacker.

Attack B

This attack involves the API to reserve seats ('/api/v1/preserveotherservice/preserveOther'). In this API, a request body comprising details of the order to be reserved is passed. Amongst the various details, the request specifies the type of seat to be reserved e.g. First Class, Second Class,

[15] https://www.selenium.dev/documentation/webdriver/
[16] https://docs.mitmproxy.org/stable/
[17] https://github.com/FudanSELab/train-ticket/blob/master/ts-consign-service/src/main/java/consign/service/ConsignServiceImpl.java#L47

Hard Seat, Soft Seat etc. In its implementation[18], there is no validation logic for the seat type. In this attack, we intercepts and modifies API requests to tamper with the "seatType" field in the reservation request payload by replacing the seatType value with another valid but different seat type (e.g., switching between First Class and Second Class). This allows the attacker to manipulate seat allocation logic, leading to inconsistencies in the booking process.

4.2.3 Dataset Statistics

The scripts are executed multiple times to produce logs for both training and test dataset.

	Training Dataset	**Test Dataset**
Number of Operations	25	25
Number of APIs targeted	48	14
Number of Attacks	0	25
Number of Logs generated	168, 799	6,054

Table 3: Dataset Statistics

The statistics regarding the number of logs and attacks collected are shown in Table 3 above. For the training dataset, 25 operations have been identified and scripted for. These operations involve 48 different API endpoints, each of which generates HTTP request/response pair logs. This resulted in a large amount of 168,799 logs. For the test dataset, 25 types of attacks have been identified and scripted for. Each type of attack is run multiple times, totalling 125 attack instances in the 6,054 logs generated.

[18] https://github.com/FudanSELab/train-ticket/blob/master/ts-preserve-other-service/src/main/java/preserveOther/service/PreserveOtherServiceImpl.java#L46

4.3. Evaluation Methodology

This section steps through the high-level methodology designed to evaluate our approach.

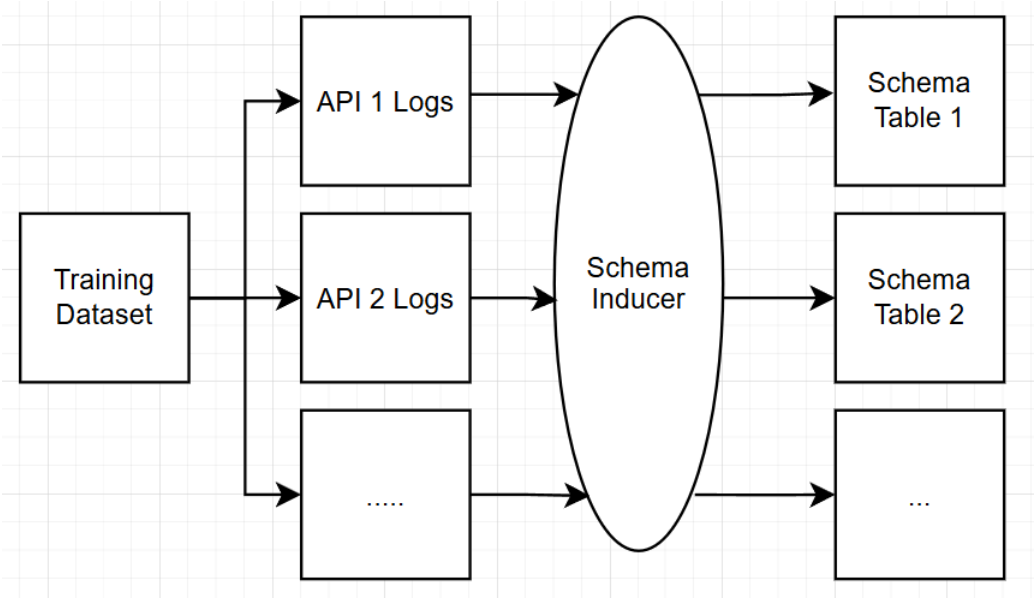

Figure 10: Converting Logs to Schema Tables

Using the training dataset, logs are first grouped by their respective API endpoints. For each API endpoint, the collected logs are processed by the schema inducer to generate its schema and a corresponding table storing the log instances.

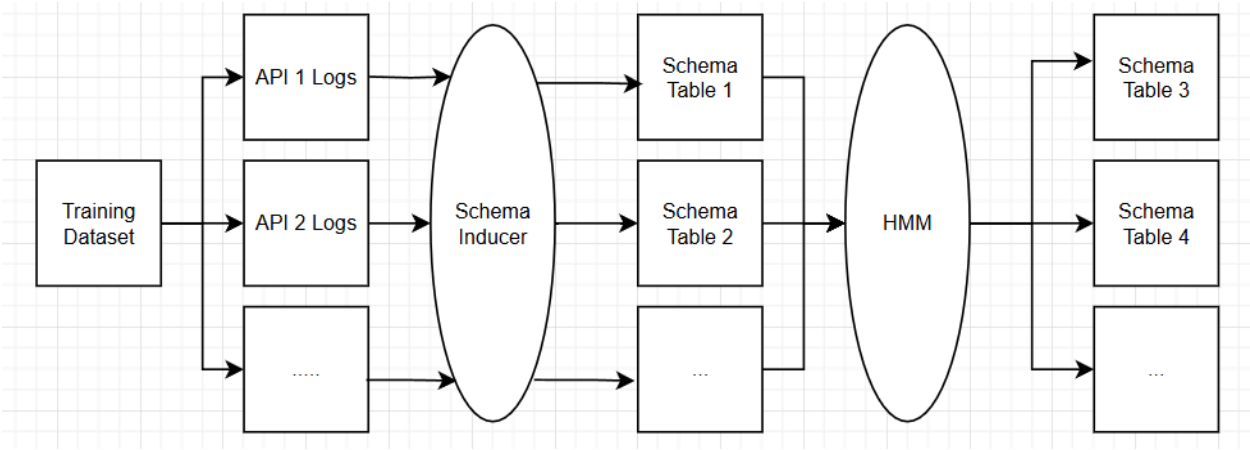

Figure 11: Grouping related Table using HMM

In addition to schema induction, a HMM is trained using the training dataset. Once trained, the HMM is used to identify API pairs that are closely related. For each of these API pairs, the corresponding log instances are joined as discussed in section 4.1.4, producing a new combined table that captures interactions between related APIs. This step enables the identification of patterns that span across multiple API calls rather than being confined to single API interactions.

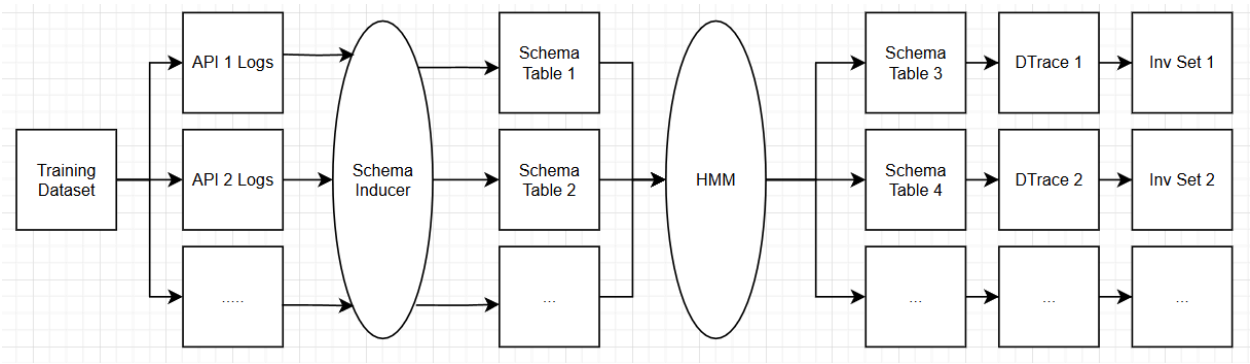

Figure 12: Generating invariants

Each of these tables is then converted into a DTrace file, which is subsequently processed by Daikon for invariant generation. The invariants extracted by Daikon are outputted as Java code and used to generate InvCheckers classes. These classes encapsulate the invariants and will later be applied to test logs to validate whether they conform to expected behavioral patterns.

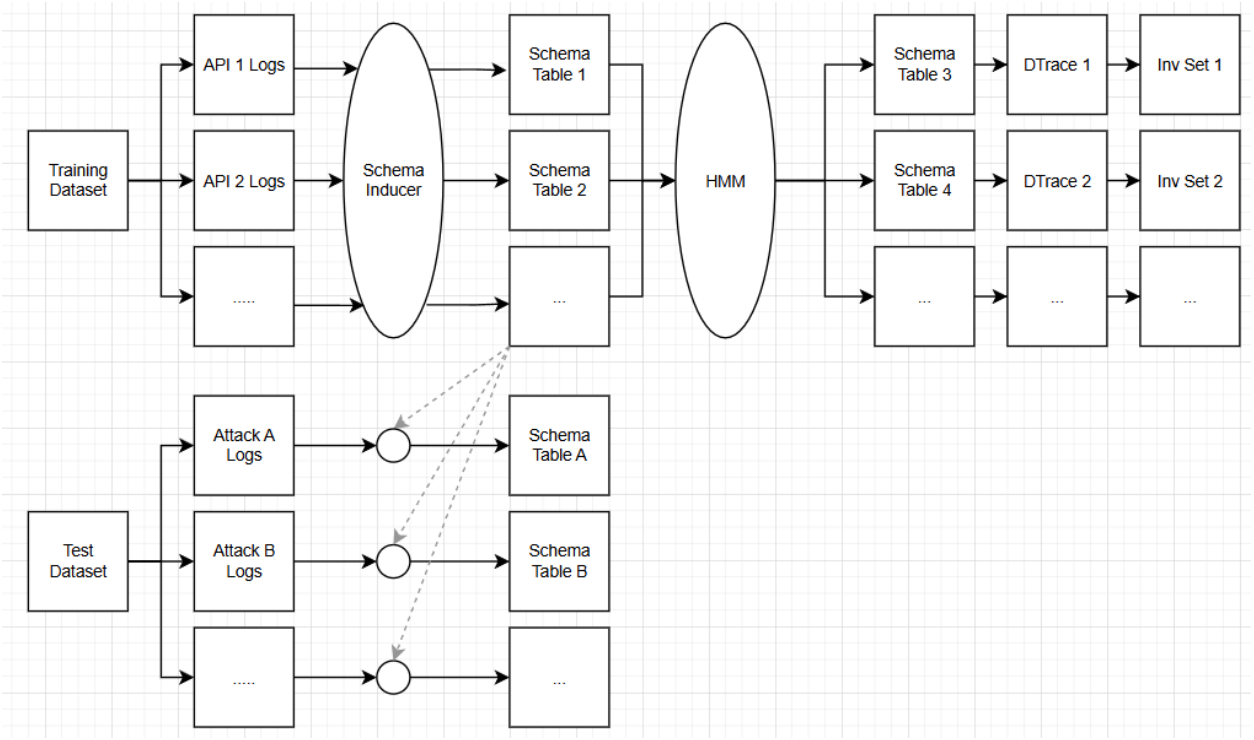

Figure 13: Converting Attack Logs to Schema Tables

When evaluating the test dataset, logs are split into individual attack scenarios and converted into their respective schema tables. Unlike in the training phase, where schemas are induced from logs, the test dataset reuses the schemas established earlier, ensuring consistency in the schemas. This prevents schema variations from influencing the results and allows a direct comparison between normal and attack scenarios.

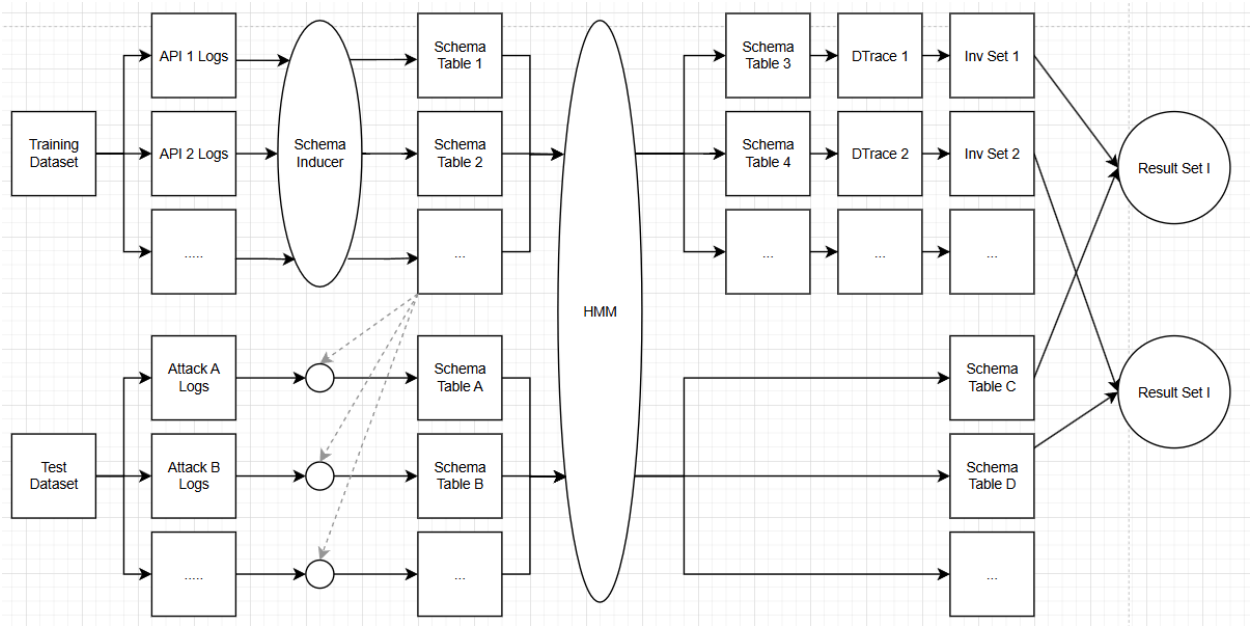

Figure 14: Evaluating test dataset

Following schema conversion, related attack log instances are joined to create new combined tables, mirroring the process used in training. Note that each table corresponds to an InvCheckers class generated during training. Lastly, the JSON representation of a table is passed to its respective InvCheckers class, where the invariant checks are performed. The result of the invariant checks are outputted for every API and API pair, under each attack.

4.4. Results

After carrying out the evaluation methodology described in section 4.3, we obtained a list of invariant checks results involving various API/API pairs for the 25 types of attacks done on Train-Ticket. In this section, we will share the overall results and analyse 2 cases to gain deeper insights.

4.4.1 Statistics

In table 4 below, we have collated the number of useful invariants that were violated during each attack run. A violated invariant is classified as useful if it represents the attack action. Depending on the attack, there are various ways in which the attack action may be represented. They commonly involve the specific modified fields or comparison with previous related API. For each attack, we provide a representative useful invariant (if any) to illustrate our classification.

Attack Type	# of useful invariants violated	Representative useful invariant	Attack identified?
0	6	**API1**: /api/v1/cancelservice/cancel/refound **API2**: /api/v1/orderservice/order/refresh **Invariant**: api2.req.loginId.equals(api1.env.user_id)	True
1	0	-	False
2	6	**API1**: /api/v1/executeservice/execute/collected **API2**: /api/v1/orderOtherService/orderOther/refresh **Invariant**: api2.req.loginId.equals(api1.env.user_id)	True
3	0	-	False
4	6	**API1**: /api/v1/executeservice/execute/execute **API2**: /api/v1/orderOtherService/orderOther/refresh **Invariant**: api2.req.loginId.equals(api1.env.user_id)	True
5	0	-	False
6	2	**API**: /api/v1/consignservice/consigns/account **Invariant**: req.user_id.equals(env.user_id)	True
7	2	**API**: /api/v1/contactservice/contacts/account	True

		Invariant: req.user_id.equals(env.user_id)	
8	4	**API**: /api/v1/orderservice/order/refresh **Invariant**: req.loginId.equals(env.user_id)	True
9	4	**API**: /api/v1/orderservice/order/refresh **Invariant**: req.loginId.equals(env.user_id)	True
10	4	**API**: /api/v1/orderservice/order/refresh **Invariant**: req.loginId.equals(env.user_id)	True
11	8	**API1**:/api/v1/consignservice/consigns **API2**: /api/v1/orderservice/order/refresh **Invariant**: api2.req.loginId.equals(api1.req.accountId)	True
12 (Attack A)	2	**API1**:/api/v1/consignservice/consigns **API2**: /api/v1/orderservice/order/refresh **Invariant**: daikon.Quant.memberOf(api1.req.orderId, api2.response.data[].id)	True
13	2	**API1**:/api/v1/consignservice/consigns **API2**: /api/v1/orderservice/order/refresh **Invariant**: daikon.Quant.memberOf(api1.req.from, api2.response.data[].from)	True
14	0	-	False
15	6	**API1**:/api/v1/inside/pay/service/inside/payment **API2**: /api/v1/orderservice/order/refresh **Invariant**: api2.req.loginId.equals(api1.env.user_id)	True
16	0	-	False
17	2	**API1**:/api/v1/rebookservice/rebook **API2**: /api/v1/orderservice/order/refresh **Invariant**: daikon.Quant.memberOf(api1.req.orderId, api2.response.data[].id)	True
18	0	-	False
19	0	-	False
20	2	**API1**:/api/v1/rebookservice/rebook **API2**: /api/v1/orderservice/order/refresh **Invariant**: daikon.Quant.memberOf(api1.req.oldTripId, api2.response.data[].trainNumber)	True

21	4	**API1**: /api/v1/contactservice/contacts/account **API2**: /api/v1/users/login **Invariant**: api2.response.data.userId.equals(api1.req.userId)	True
22	4	**API1**:/api/v1/preserveotherservice/preserveOther **API2**: /api/v1/users/login **Invariant**: api2.response.data.userId.equals(api1.req.accountId)	True
23	1	**API**: /api/v1/preserveotherservice/preserveOther **Invariant**: api.req.assurance.equals("0") || api.req.assurance.equals("1")	True
24 (Attack B)	0	-	False

Table 4: Results for all attacks

At the attack level, true positives are attacks that are identified by meaningful invariants correctly, while attacks with only non-meaningful invariants give false positives. False negatives are attacks that are not identified, otherwise non-attacks that are classified as an attack are true negatives. Given these, we achieve the following recall and precision with comparison to WebNorm:

	Recall	**Precision**
WebNorm	0.704	1.000
Our approach	1.000	0.680

Table 5: Statistics for our optimised approach vs WebNorm

4.4.2 Case Analysis

In this section, we will analyse the 2 attacks illustrated in section 4.2.2. Attack A is attack type 12, while Attack B is attack type 24.

Attack A/Attack Type 12

In this attack, as the field “orderId” is tampered with, we expect an invariant that describes the validity of the “orderId” field.

Amongst the various violated invariants, we identified a representative useful invariant that corresponds to our expectation as shown in table 4. This invariant involves 2 API, /api/v1/consignservice/consigns and /api/v1/orderservice/order/refresh, and is literally defined as “daikon.Quant.memberOf(api1.req.orderId, api2.response.data[].id)". In other words, this invariant states that the “orderId” field in the request payload of /api/v1/consignservice/consigns API, must be found in the list of orders returned by the /api/v1/orderservice/order/refresh API previously.

This invariant is intuitive and represents a common use case in web application; train-ticket first calls /api/v1/orderservice/order/refresh API to get the list of orders belonging to the user, display them and allow the user to choose one of the orders to place consignment order for. Thus, the orderId in the consignment order should be present in the list of orders returned from /api/v1/orderservice/order/refresh API. Any violation could indicate that the user did not conform to this use case, and could have instead tampered with the request.

Furthermore, this invariant represents an event relation between the 2 APIs, as correctly identified by the use of HMM discussed in section 4.1.3.

Attack B/Attack Type 24

To recap, this attack involves modifying the seat type of a seat in an order request. Lets hypothesize the invariants that will allow us to identify such an attack.

Firstly, a simple invariant that checks for the validity of the seat type (e.g. seatType ∈ [first_class, second_class, …]) is not sufficient. This is because the modified seat type value is still one of the valid seat types.

To identify that the seat type is inconsistent with the specified seat, we need a more complex invariant that validates the seatType in the context of the specific seat being referenced. In particular, a meaningful invariant could be:
querySeats.response.data[i].id == orderTickets.request.seatId &&
querySeats.response.data[i].seatType == orderTickets.request.seatType.
In other words, for the selected seat, the seatType in the order request must match the seatType returned from the querySeats API. This invariant ensures that the seatType provided in the order request has not been tampered with and is consistent with the state of the web application.

However, there is a limitation in how Daikon infers such invariants. Due to the way DTrace flattens object arrays, fields belonging to the same logical object (such as seatId and seatType within an order) are recorded as separate arrays of primitive values. Daikon, being unaware of the grouping semantics, cannot infer object-level invariants—such as linking a specific seatId to its seatType—even though they logically belong together. Thus, no such invariants are deduced during training, and this attack is not identified during testing.

5. Conclusion

In this report, we propose an approach to generate invariants through non-LLM means in order to optimize WebNorm for industrial use. From our evaluation results, we can conclude that this approach can successfully generate meaningful invariants that helps to identify certain attacks. However, the number of false positives are high as indicated by the precision score, which indicates that further work is required. Regardless, noting a higher recall score, this approach performs comparably to WebNorm and provides a promising direction for future work.

5.1. Future works

Given our current analysis, future works could focus on addressing the current limitation of this approach. In particular, we could consider improving on our approach to identify related events and to infer more complex invariants.

5.1.1 Events Relations

In our approach, we group related logs through the use of HMM. Given that our initial goal was to achieve a healthy recall score, we focused on generating all potential related events. This was achieved by adjusting the threshold set on the HMM. As a result, the accuracy of the HMM prediction is not our focus, which eventually led to too many relations being identified. Subsequently, many APIs and logs were combined which caused Daikon to output many incorrect invariants.

Thus, future work can look into improving the accuracy of the HMM. Furthermore, events relations can be extended beyond the current dimension of 2 API. Multiple APIs can be associated in arbitrary events, and grouping their logs will enable more invariants between these APIs to be identified.

5.1.2 Complex Invariants

With reference to the case analysis on Attack B in section 4.4.2, complex invariants are not deduced due to our approach in DTrace generation and Daikon internals. On a high-level, Daikon has a list of predefined invariants that it uses to check against the trace records before deciding on the output. These invariants represent common relationships observed between variables.

Thus, we could look at extending daikon to support our DTrace file that handles complex object arrays differently. In particular, we can define and declare new invariants for Daikon to check. These new invariants should represent common relationships between flatten arrays with a common parent variable.

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Appendix

Appendix A: Example DTrace File

```
// Metadata
decl-version 2.0

// Declarations
ppt Main.main():::Point
    ppt-type exit
    variable status
        var-kind variable
        dec-type int
        rep-type int
    variable msg
        var-kind variable
        dec-type java.lang.String
        rep-type java.lang.String
    variable orders
        var-kind variable
        dec-type java.lang.Object[]
        rep-type hashcode
    variable orders[..].id
        var-kind array
        enclosing-var orders
        array 1
        dec-type java.lang.String[]
        rep-type java.lang.String[]
    variable orders[..].price
        var-kind array
        enclosing-var orders
        array 1
        dec-type double[]
        rep-type double[]
```

```
// Trace Records
Main.main():::Point
status
1
1
msg
"Query Orders For Refresh Success"
1
orders
12345
1
orders[..].id
["a100" "b101"]
1
orders[..].price
[100.00 200.00]
1
```

Appendix B: Schema Inducer Data Type

Data Type Name	Properties Tracked
String	- Is Nullable? - Minimum length - Maximum length - Is unique?
Int	- Is Nullable? - Minimum Value - Maximum Value - Is unique?
Float	
Bool	
Bytes	- Is Nullable? - Minimum length - Maximum length
Array	- Is Nullable? - Minimum length - Maximum length - Element schema
Object	- Is Nullable? - Minimum length - Maximum length - Minimum Value - Maximum Value - Fields - Is unique?
Dict	
Null	
Unknown	

Appendix C: InvariantChecker

InvariantChecker.java

```
##############################################################
package daikon_inv_gen;

import java.io.ByteArrayOutputStream;
import java.io.File;
import java.io.IOException;
import java.io.PrintWriter;
import java.net.URL;
import java.net.URLClassLoader;
import java.util.ArrayList;
import java.util.List;

import lombok.AllArgsConstructor;
import lombok.Getter;
import lombok.extern.slf4j.Slf4j;

@Slf4j
public class InvariantChecker {

   public static String getStackTrace(Throwable theException) {
      {{ redacted as it is not relevant to core logic }}
   }

   public InvariantCheckResult checkInvariant(String dataFile, String invClassPath, String
invClassName) {
      try {
         return checkInvariantInner(dataFile, invClassPath, invClassName);
      } catch (Exception e) {
         log.error("Error while checking invariants", e);
```

```
        return new InvariantCheckResult(false, getStackTrace(e), null);
    }
  }

  public InvariantCheckResult checkInvariantInner(String dataFile, String invClassPath, String
invClassName)
        throws Exception {
    DataLoader loader = new DataLoader();
    TableData data = loader.loadData(dataFile);

    ArrayList<FailedInvariantCheck> failedInvariantChecks = new ArrayList<>();
    boolean result = true;

    URL url = new File(invClassPath).toURI().toURL();
    try (URLClassLoader classLoader = new URLClassLoader(new URL[] { url })) {
        Class<?> invClass = classLoader.loadClass(invClassName);

        InvCheckerAll invChecker = (InvCheckerAll)
invClass.getDeclaredConstructor().newInstance();
        InvChecker[] checkers = invChecker.getInvCheckers();

        for (int rowIndex = 0; rowIndex < data.getSize(); rowIndex++) {
            for (int invariantIndex = 0; invariantIndex < checkers.length; invariantIndex++) {
                try {
                    if (!checkers[invariantIndex].check(data.getVariablesAt(rowIndex))) {
                        // log.info("Invariant idx {} failed on row {}", invariantIndex, rowIndex);
                        failedInvariantChecks.add(new FailedInvariantCheck(invariantIndex,
rowIndex, null));
                        result = false;
                    }
                } catch (Exception e) {
```

```
                log.error("Error while checking invariant idx {} on row {}.", invariantIndex,
rowIndex, e);
                failedInvariantChecks.add(new FailedInvariantCheck(invariantIndex, rowIndex,
getStackTrace(e)));
                result = false;
            }
        }
    }
  }

  return new InvariantCheckResult(result, null, failedInvariantChecks);
}

@Getter
@AllArgsConstructor
public static class InvariantCheckResult {
  private boolean result;
  private String exception;
  private List<FailedInvariantCheck> failedInvariantChecks;
}

@Getter
@AllArgsConstructor
public static class FailedInvariantCheck {
  private int invariantIndex;
  private int rowIndex;
  private String exception;
}
}
```

##

Main.java

```
##############################################################
import java.io.FileWriter;
import java.util.concurrent.Callable;

import com.google.gson.Gson;
import com.google.gson.GsonBuilder;

import daikon_inv_gen.InvariantChecker.InvariantCheckResult;
import lombok.extern.slf4j.Slf4j;
import picocli.CommandLine;
import picocli.CommandLine.IExitCodeExceptionMapper;
import picocli.CommandLine.Parameters;

@Slf4j
public class Main implements Callable<Integer>, IExitCodeExceptionMapper {
   @Parameters(index = "0")
   private String dataFile;
   @Parameters(index = "1")
   private String invClassPath;
   @Parameters(index = "2")
   private String invClassName;
   @Parameters(index = "3")
   private String outputFileName;

   public static void main(String[] args) {
      int exitCode = new CommandLine(new Main()).execute(args);
      System.exit(exitCode);
   }

   @Override
```

```
    public int getExitCode(Throwable exception) {
        log.error("Error while checking invariants", exception);
        return 2;
    }

    @Override
    public Integer call() throws Exception {
        InvariantChecker checker = new InvariantChecker();
        InvariantCheckResult result = checker.checkInvariant(dataFile, invClassPath,
invClassName);
        GsonBuilder builder = new GsonBuilder();
        builder = builder.setPrettyPrinting();
        builder = builder.serializeNulls();
        Gson gson = builder.create();
        try (FileWriter writer = new FileWriter(outputFileName)) {
            gson.toJson(result, writer);
        }
        if (result.isResult()) {
            return 0;
        } else {
            return 1;
        }
    }
}
```

##

Appendix D: Types of Attack

Attack Type	Target API	Attack Action	Potential Impact
0	/api/v1/cancelservice/cancel/refound	The attacker intercepts and modifies API requests to impersonate the target user (to_user) by replacing the loginId field in the request payload.	The attacker cancels orders belonging to another user without their consent
1	/api/v1/cancelservice/cancel	The attacker intercepts and modifies the API request to tamper with the order_id in the request path, replacing it with an unauthorized order_id.	The attacker cancels an order that they are not authorized to cancel, potentially affecting another user's order.
2	/api/v1/executeservice/execute/collected	The attacker intercepts and modifies API requests to impersonate the target user (to_user) by replacing the loginId field in the request payload with the target user's loginId.	The attacker collects tickets belonging to another user without their consent.
3	/api/v1/executeservice/execute/collected	The attacker intercepts and modifies the API request to tamper with the order_id in the request path, replacing it with an unauthorized order_id that the attacker is not permitted to collect.	The attacker collects a ticket for an order that they are not authorized to collect, potentially affecting another user's ticket.
4	/api/v1/executeservice/execute/execute	The attacker intercepts and modifies API requests to impersonate the target user (to_user) by replacing the loginId field in the request payload with the target user's loginId.	The attacker gains unauthorized access to enter the station using another user's ticket.
5	/api/v1/executeservice/execute/execute	The attacker intercepts and modifies the API request to tamper with the order_id in the request path, replacing it with an unauthorized order_id that the attacker is not permitted to use for entering the station.	The attacker gains unauthorized access to enter the station using a tampered order_id, potentially bypassing ticket validation.
6	/api/v1/consignservice/consigns/	The attacker intercepts and modifies the API request to tamper with the user_id in the request path,	The attacker gains unauthorized access to view consignments associated

	account	replacing it with another user's user_id.	with another user's account.
7	/api/v1/contactservice/contacts/account	The attacker intercepts and modifies the API request to tamper with the user_id in the request path, replacing it with another user's user_id.	The attacker gains unauthorized access to view contact details associated with another user's account.
8	/api/v1/orderservice/order/refresh	The attacker intercepts and modifies API requests to impersonate the target user (to_user) by replacing the loginId field in the request payload with the target user's loginId.	The attacker gains unauthorized access to view order details associated with another user's account.
9	/api/v1/orderservice/order/refresh	The attacker intercepts and modifies API requests to impersonate the target user (to_user) by replacing the loginId field in the request payload with the target user's loginId.	The attacker gains unauthorized access to view station-related details associated with another user's account.
10	/api/v1/orderservice/order/refresh	The attacker intercepts and modifies API requests to impersonate the target user (to_user) by replacing the loginId field in the request payload with the target user's loginId.	The attacker gains unauthorized access to view ticket-related details associated with another user's account.
11	/api/v1/consignservice/consigns	The attacker intercepts and modifies API requests to impersonate the target user (to_user) by replacing the loginId field in the request payload with the target user's loginId.	The attacker gains unauthorized access to order consignments using another user's account, potentially leading to financial or logistical consequences for the victim.
12	/api/v1/consignservice/consigns	The attacker intercepts and modifies API requests to tamper with the orderId field in the request payload. The original orderId is replaced with a randomly selected orderId from a list of orders that do not belong to the user.	The attacker gains unauthorized access to place consignment orders for orders that do not belong to them, potentially causing financial or logistical issues for the legitimate owners of the orders.
13	/api/v1/consignservice/consigns	The attacker intercepts and modifies API requests to tamper with the from or to station fields in	The attacker gains unauthorized access to modify consignment orders

		the request payload. The original station is replaced with a randomly selected station that is not part of the original consignment order.	by changing the from or to station, potentially causing logistical issues or misrouting of consignments.
14	/api/v1/inside/pay/service/inside/payment	The attacker intercepts and modifies API requests to tamper with the userId field in the payment request payload. The original userId is replaced with another user's userId.	The attacker gains unauthorized access to make payments on behalf of another user, potentially leading to financial loss for the victim.
15	/api/v1/inside/pay/service/inside/payment	The attacker intercepts and modifies API requests to impersonate the target user (to_user) by replacing the loginId field in the request payload with the target user's loginId.	The attacker gains unauthorized access to pay for orders associated with another user's account, potentially causing financial loss or unauthorized transactions for the victim.
16	/api/v1/inside/pay/service/inside/payment	The attacker intercepts and modifies API requests to tamper with the tripId field in the payment request payload. The original tripId is replaced with a randomly selected tripId from a list of trips that do not belong to the user.	The attacker gains unauthorized access to pay for trips that do not belong to them, potentially causing financial loss or unauthorized transactions for the victim.
17	/api/v1/rebookservice/rebook	The attacker intercepts and modifies API requests to tamper with the orderId and oldTripId fields in the rebooking request payload. The original orderId and oldTripId are replaced with values belonging to another user.	The attacker gains unauthorized access to rebook orders for trips associated with another user, potentially causing disruptions to the victim's travel plans or financial loss.
18	/api/v1/rebookservice/rebook	The attacker intercepts and modifies API requests to tamper with the loginId field in the rebooking request payload. The original loginId is replaced with another user's loginId.	The attacker gains unauthorized access to rebook orders for another user, potentially causing disruptions to the victim's travel plans or financial loss.
19	/api/v1/rebookservice/rebook	The attacker intercepts and modifies API requests to tamper with the orderId field in the rebooking request payload. The attacker replaces the orderId with an order that has a status that does	The attacker exploits the system to bypass restrictions on rebooking, potentially causing disruptions or inconsistencies in the order management system.

		not allow rebooking.	
20	/api/v1/rebookservice/rebook	The attacker intercepts and modifies API requests to tamper with the oldTripId field in the rebooking request payload. The attacker replaces the oldTripId with a train trip ID that does not belong to the user or is invalid.	The attacker exploits the system to bypass restrictions on rebooking, potentially causing disruptions or inconsistencies in the order management system, or allowing unauthorized rebooking of trips.
21	/api/v1/contactservice/contacts/account	The attacker intercepts and modifies API requests to tamper with the path_param_0 in the request URL, replacing it with another user's user_id.	The attacker gains unauthorized access to another user's contact information, potentially leading to privacy violations, unauthorized bookings, or misuse of the victim's account.
22	/api/v1/preserveotherservice/preserveOther	The attacker intercepts and modifies API requests to tamper with the accountId field in the reservation request payload. The attacker replaces the accountId with another user's user_id.	The attacker gains unauthorized access to reserve tickets using another user's account, potentially leading to financial loss, unauthorized bookings, or misuse of the victim's account.
23	/api/v1/preserveotherservice/preserveOther	The attacker intercepts and modifies API requests to tamper with the assurance field in the reservation request payload. The attacker replaces the assurance value with a random invalid or unexpected value (e.g., outside the valid range)..	The attacker exploits the system to bypass or manipulate assurance-related logic, potentially causing errors, unexpected behavior, or security vulnerabilities in the reservation process.
24	/api/v1/preserveotherservice/preserveOther	Execution: The attacker intercepts and modifies API requests to tamper with the seatType field in the reservation request payload. The attacker replaces the seatType value with another valid but incorrect seat type (e.g., switching between "2" and "3").	The attacker exploits the system to bypass or manipulate seat allocation logic, potentially leading to incorrect seat assignments, reservation errors, or inconsistencies in the booking process.

Appendix E0: Attack 0 Evaluation output

```
# Attack 0
## _api_v1_cancelservice_cancel_refound@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_36)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_36: log_data.env.is_user
- 113: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_37) > 0
   - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
   - webnorm_invariant_gen_var_37: log_data.env.user_id
## _api_v1_cancelservice_cancel_refound@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_14)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_14: log_data.env.is_user
## _api_v1_cancelservice_cancel_refound@@_api_v1_orderOtherService_orderOther_refresh
- 5: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_39)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_39: log_data.env.user_id
## _api_v1_cancelservice_cancel_refound@@_api_v1_orderservice_order_refresh
- 5: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_39)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_39: log_data.env.user_id
## _api_v1_cancelservice_cancel_refound@@_api_v1_travel2service_trips_left
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_29)
   - webnorm_invariant_gen_var_23: extra@@log_data.response.status
   - webnorm_invariant_gen_var_29: log_data.env.is_user
## _api_v1_orderOtherService_orderOther_refresh
- 3: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_12)
   - webnorm_invariant_gen_var_6: log_data.arguments.loginId
   - webnorm_invariant_gen_var_12: log_data.env.user_id
##
_api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderOtherService_orderOther_re
fresh
- 9: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_46: log_data.env.user_id
## _api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderservice_order_refresh
- 9: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_45)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_45: log_data.env.is_user
## _api_v1_orderservice_order_refresh
- 3: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_12)
   - webnorm_invariant_gen_var_6: log_data.arguments.loginId
   - webnorm_invariant_gen_var_12: log_data.env.user_id
```

_api_v1_orderservice_order_refresh@@_api_v1_orderservice_order_refresh
- 9: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_46: log_data.env.user_id

Appendix E1: Attack 1 Evaluation output

```
# Attack 1
## _api_v1_cancelservice_cancel
- 0: webnorm_invariant_gen_var_2.equals(webnorm_invariant_gen_var_6)
   - webnorm_invariant_gen_var_2: log_data.arguments.path_param_1
   - webnorm_invariant_gen_var_6: log_data.env.user_id
## _api_v1_cancelservice_cancel@@_api_v1_cancelservice_cancel
- 3: webnorm_invariant_gen_var_2.equals(webnorm_invariant_gen_var_18)
   - webnorm_invariant_gen_var_2: extra@@log_data.arguments.path_param_1
   - webnorm_invariant_gen_var_18: log_data.env.user_id
## _api_v1_cancelservice_cancel@@_api_v1_cancelservice_cancel_refound
- 0: webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_12)
   - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
   - webnorm_invariant_gen_var_12: log_data.arguments.path_param_0
- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_16)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_16: log_data.env.is_user
- 3: webnorm_invariant_gen_var_5.equals(webnorm_invariant_gen_var_13)
   - webnorm_invariant_gen_var_5: extra@@log_data.env.user_id
   - webnorm_invariant_gen_var_13: log_data.arguments.path_param_1
- 4: webnorm_invariant_gen_var_5.equals(webnorm_invariant_gen_var_17)
   - webnorm_invariant_gen_var_5: extra@@log_data.env.user_id
   - webnorm_invariant_gen_var_17: log_data.env.user_id
## _api_v1_cancelservice_cancel@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_37)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_37: log_data.env.is_user
- 113: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_33) > 0
   - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
   - webnorm_invariant_gen_var_33: log_data.arguments.path_param_0
- 114: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_34) > 0
   - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
   - webnorm_invariant_gen_var_34: log_data.arguments.path_param_1
## _api_v1_cancelservice_cancel@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_15)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_15: log_data.env.is_user
## _api_v1_cancelservice_cancel@@_api_v1_foodservice_foods
- 46: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_33) > 0
   - webnorm_invariant_gen_var_3: extra@@log_data.arguments.path_param_2
   - webnorm_invariant_gen_var_33: log_data.arguments.path_param_1
## _api_v1_cancelservice_cancel@@_api_v1_orderOtherService_orderOther_refresh
- 5: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_36)
```

- webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
- webnorm_invariant_gen_var_36: log_data.arguments.path_param_1

_api_v1_cancelservice_cancel@@_api_v1_orderservice_order_refresh

- 5: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_36: log_data.arguments.path_param_1

_api_v1_cancelservice_cancel@@_api_v1_travel2service_trips_left

- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_30)
  - webnorm_invariant_gen_var_23: extra@@log_data.response.status
  - webnorm_invariant_gen_var_30: log_data.env.is_user
- 38: webnorm_invariant_gen_var_2.compareTo(webnorm_invariant_gen_var_26) > 0
  - webnorm_invariant_gen_var_2: extra@@log_data.arguments.endPlace
  - webnorm_invariant_gen_var_26: log_data.arguments.path_param_0
- 39: webnorm_invariant_gen_var_2.compareTo(webnorm_invariant_gen_var_27) > 0
  - webnorm_invariant_gen_var_2: extra@@log_data.arguments.endPlace
  - webnorm_invariant_gen_var_27: log_data.arguments.path_param_1

_api_v1_cancelservice_cancel@@_api_v1_travelservice_trips_left

- 38: webnorm_invariant_gen_var_2.compareTo(webnorm_invariant_gen_var_26) > 0
  - webnorm_invariant_gen_var_2: extra@@log_data.arguments.endPlace
  - webnorm_invariant_gen_var_26: log_data.arguments.path_param_0
- 39: webnorm_invariant_gen_var_2.compareTo(webnorm_invariant_gen_var_27) > 0
  - webnorm_invariant_gen_var_2: extra@@log_data.arguments.endPlace
  - webnorm_invariant_gen_var_27: log_data.arguments.path_param_1

_api_v1_cancelservice_cancel@@_api_v1_users_login

- 2: webnorm_invariant_gen_var_11.equals(webnorm_invariant_gen_var_18)
  - webnorm_invariant_gen_var_11: extra@@log_data.response.data.userId
  - webnorm_invariant_gen_var_18: log_data.arguments.path_param_1

_api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderservice_order_refresh

- 9: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_45)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_45: log_data.env.is_user

Appendix E2: Attack 2 Evaluation output

```
# Attack 2
## _api_v1_executeservice_execute_collected@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_36)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_36: log_data.env.is_user
- 113: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_37) > 0
   - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
   - webnorm_invariant_gen_var_37: log_data.env.user_id
## _api_v1_executeservice_execute_collected@@_api_v1_consignservice_consigns_account
- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_26)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_26: log_data.env.is_user
## _api_v1_executeservice_execute_collected@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_14)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_14: log_data.env.is_user
## _api_v1_executeservice_execute_collected@@_api_v1_contactservice_contacts_account
- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_21)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_21: log_data.env.is_user
## _api_v1_executeservice_execute_collected@@_api_v1_foodservice_foods
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_7, webnorm_invariant_gen_var_35)
   - webnorm_invariant_gen_var_7: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_35: log_data.env.is_user
- 44: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_32) > 0
   - webnorm_invariant_gen_var_3: extra@@log_data.arguments.path_param_2
   - webnorm_invariant_gen_var_32: log_data.arguments.path_param_0
##
_api_v1_executeservice_execute_collected@@_api_v1_orderOtherService_orderOther_refres
h
- 5: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_39)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_39: log_data.env.user_id
## _api_v1_executeservice_execute_collected@@_api_v1_orderservice_order_refresh
- 5: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_39)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_39: log_data.env.user_id
## _api_v1_executeservice_execute_collected@@_api_v1_travel2service_trips_left
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_29)
   - webnorm_invariant_gen_var_23: extra@@log_data.response.status
   - webnorm_invariant_gen_var_29: log_data.env.is_user
## _api_v1_executeservice_execute_collected@@_api_v1_travelservice_trips_left
```

- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_29)
  - webnorm_invariant_gen_var_23: extra@@log_data.response.status
  - webnorm_invariant_gen_var_29: log_data.env.is_user

_api_v1_orderOtherService_orderOther_refresh
- 3: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_12)
  - webnorm_invariant_gen_var_6: log_data.arguments.loginId
  - webnorm_invariant_gen_var_12: log_data.env.user_id

_api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderOtherService_orderOther_refresh
- 9: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_46: log_data.env.user_id

_api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderservice_order_refresh
- 9: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_45)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_45: log_data.env.is_user

_api_v1_orderservice_order_refresh
- 3: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_12)
  - webnorm_invariant_gen_var_6: log_data.arguments.loginId
  - webnorm_invariant_gen_var_12: log_data.env.user_id

_api_v1_orderservice_order_refresh@@_api_v1_orderservice_order_refresh
- 9: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_46: log_data.env.user_id

Appendix E3: Attack 3 Evaluation output

```
# Attack 3
## _api_v1_executeservice_execute_collected@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_36)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_36: log_data.env.is_user
## _api_v1_executeservice_execute_collected@@_api_v1_consignservice_consigns_account
- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_26)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_26: log_data.env.is_user
## _api_v1_executeservice_execute_collected@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_14)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_14: log_data.env.is_user
## _api_v1_executeservice_execute_collected@@_api_v1_contactservice_contacts_account
- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_21)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_21: log_data.env.is_user
## _api_v1_executeservice_execute_collected@@_api_v1_foodservice_foods
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_7, webnorm_invariant_gen_var_35)
   - webnorm_invariant_gen_var_7: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_35: log_data.env.is_user
- 44: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_32) > 0
   - webnorm_invariant_gen_var_3: extra@@log_data.arguments.path_param_2
   - webnorm_invariant_gen_var_32: log_data.arguments.path_param_0
## _api_v1_executeservice_execute_collected@@_api_v1_travel2service_trips_left
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_29)
   - webnorm_invariant_gen_var_23: extra@@log_data.response.status
   - webnorm_invariant_gen_var_29: log_data.env.is_user
## _api_v1_executeservice_execute_collected@@_api_v1_travelservice_trips_left
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_29)
   - webnorm_invariant_gen_var_23: extra@@log_data.response.status
   - webnorm_invariant_gen_var_29: log_data.env.is_user
## _api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderservice_order_refresh
- 9: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_45)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_45: log_data.env.is_user
```

Appendix E4: Attack 4 Evaluation output

```
# Attack 4
## _api_v1_executeservice_execute_execute@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_36)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_36: log_data.env.is_user
## _api_v1_executeservice_execute_execute@@_api_v1_consignservice_consigns_account
- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_26)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_26: log_data.env.is_user
## _api_v1_executeservice_execute_execute@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_14)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_14: log_data.env.is_user
## _api_v1_executeservice_execute_execute@@_api_v1_contactservice_contacts_account
- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_21)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_21: log_data.env.is_user
## _api_v1_executeservice_execute_execute@@_api_v1_foodservice_foods
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_7, webnorm_invariant_gen_var_35)
   - webnorm_invariant_gen_var_7: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_35: log_data.env.is_user
##
_api_v1_executeservice_execute_execute@@_api_v1_orderOtherService_orderOther_refresh
- 5: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_39)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_39: log_data.env.user_id
## _api_v1_executeservice_execute_execute@@_api_v1_orderservice_order_refresh
- 5: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_39)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_39: log_data.env.user_id
## _api_v1_executeservice_execute_execute@@_api_v1_travel2service_trips_left
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_29)
   - webnorm_invariant_gen_var_23: extra@@log_data.response.status
   - webnorm_invariant_gen_var_29: log_data.env.is_user
## _api_v1_executeservice_execute_execute@@_api_v1_travelservice_trips_left
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_29)
   - webnorm_invariant_gen_var_23: extra@@log_data.response.status
   - webnorm_invariant_gen_var_29: log_data.env.is_user
## _api_v1_orderOtherService_orderOther_refresh
- 3: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_12)
   - webnorm_invariant_gen_var_6: log_data.arguments.loginId
   - webnorm_invariant_gen_var_12: log_data.env.user_id
```

_api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderOtherService_orderOther_refresh
- 9: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_46: log_data.env.user_id

_api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderservice_order_refresh
- 9: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_45)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_45: log_data.env.is_user

_api_v1_orderservice_order_refresh
- 3: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_12)
  - webnorm_invariant_gen_var_6: log_data.arguments.loginId
  - webnorm_invariant_gen_var_12: log_data.env.user_id

_api_v1_orderservice_order_refresh@@_api_v1_orderservice_order_refresh
- 9: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_46: log_data.env.user_id

Appendix E5: Attack 5 Evaluation output

```
# Attack 5
## _api_v1_executeservice_execute_execute@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_36)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_36: log_data.env.is_user
- 111: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_33) > 0
   - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
   - webnorm_invariant_gen_var_33: log_data.arguments.path_param_0
## _api_v1_executeservice_execute_execute@@_api_v1_consignservice_consigns_account
- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_26)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_26: log_data.env.is_user
## _api_v1_executeservice_execute_execute@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_14)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_14: log_data.env.is_user
## _api_v1_executeservice_execute_execute@@_api_v1_contactservice_contacts_account
- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_21)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_21: log_data.env.is_user
## _api_v1_executeservice_execute_execute@@_api_v1_foodservice_foods
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_7, webnorm_invariant_gen_var_35)
   - webnorm_invariant_gen_var_7: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_35: log_data.env.is_user
- 44: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_32) > 0
   - webnorm_invariant_gen_var_3: extra@@log_data.arguments.path_param_2
   - webnorm_invariant_gen_var_32: log_data.arguments.path_param_0
## _api_v1_executeservice_execute_execute@@_api_v1_travel2service_trips_left
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_29)
   - webnorm_invariant_gen_var_23: extra@@log_data.response.status
   - webnorm_invariant_gen_var_29: log_data.env.is_user
## _api_v1_executeservice_execute_execute@@_api_v1_travelservice_trips_left
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_29)
   - webnorm_invariant_gen_var_23: extra@@log_data.response.status
   - webnorm_invariant_gen_var_29: log_data.env.is_user
- 37: webnorm_invariant_gen_var_2.compareTo(webnorm_invariant_gen_var_26) > 0
   - webnorm_invariant_gen_var_2: extra@@log_data.arguments.endPlace
   - webnorm_invariant_gen_var_26: log_data.arguments.path_param_0
```

Appendix E6: Attack 6 Evaluation output

```
# Attack 6
## _api_v1_consignservice_consigns_account
- 0: webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_5)
   - webnorm_invariant_gen_var_1: log_data.arguments.path_param_0
   - webnorm_invariant_gen_var_5: log_data.env.user_id
## _api_v1_consignservice_consigns_account@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_36)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_36: log_data.env.is_user
## _api_v1_consignservice_consigns_account@@_api_v1_consignservice_consigns_account
- 2: webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_27)
   - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
   - webnorm_invariant_gen_var_27: log_data.env.user_id
## _api_v1_consignservice_consigns_account@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_14)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_14: log_data.env.is_user
##
_api_v1_consignservice_consigns_account@@_api_v1_orderOtherService_orderOther_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_38)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_38: log_data.env.is_user
## _api_v1_consignservice_consigns_account@@_api_v1_orderservice_order_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_38)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_38: log_data.env.is_user
```

Appendix E7: Attack 7 Evaluation output

```
# Attack 7
## _api_v1_contactservice_contacts_account
- 0: webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_5)
   - webnorm_invariant_gen_var_1: log_data.arguments.path_param_0
   - webnorm_invariant_gen_var_5: log_data.env.user_id
##
_api_v1_contactservice_contacts_account@@_api_v1_assuranceservice_assurances_types
- 4: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_8) >= 0
   - webnorm_invariant_gen_var_4: extra@@log_data.env.user_id
   - webnorm_invariant_gen_var_8: log_data.arguments.path_param_0
## _api_v1_contactservice_contacts_account@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_36)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_36: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_consignservice_consigns_account
- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_26)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_26: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_14)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_14: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_contactservice_contacts_account
- 2: webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_22)
   - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
   - webnorm_invariant_gen_var_22: log_data.env.user_id
##
_api_v1_contactservice_contacts_account@@_api_v1_orderOtherService_orderOther_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_38)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_38: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_orderservice_order_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_38)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_38: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_travelservice_trips_left
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_29)
   - webnorm_invariant_gen_var_23: extra@@log_data.response.status
   - webnorm_invariant_gen_var_29: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_users_login
- 2: webnorm_invariant_gen_var_11.equals(webnorm_invariant_gen_var_17)
   - webnorm_invariant_gen_var_11: extra@@log_data.response.data.userId
```

- webnorm_invariant_gen_var_17: log_data.arguments.path_param_0
_api_v1_foodservice_foods@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_39)
  - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_39: log_data.env.is_user
- 73: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
  - webnorm_invariant_gen_var_33: log_data.arguments.path_param_0
- 76: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_36) < 0
  - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
  - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
- 116: webnorm_invariant_gen_var_8.compareTo(webnorm_invariant_gen_var_36) < 0
  - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
  - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
- 124: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
  - webnorm_invariant_gen_var_33: log_data.arguments.path_param_0
- 127: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_36) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
  - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
- 144: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_36) < 0
  - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
  - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
- 171: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_33: log_data.arguments.path_param_0
- 174: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_36) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
_api_v1_foodservice_foods@@_api_v1_consignservice_consigns_account
- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_29)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_29: log_data.env.is_user
- 66: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_23) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_23: log_data.arguments.path_param_0
- 69: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_26) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_26: log_data.arguments.path_param_3
_api_v1_foodservice_foods@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_17)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_17: log_data.env.is_user
- 19: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_11) < 0

  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_11: log_data.arguments.path_param_0
- 22: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_14) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_14: log_data.arguments.path_param_3
_api_v1_foodservice_foods@@_api_v1_orderOtherService_orderOther_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_41)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_41: log_data.env.is_user
- 91: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_35) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_35: log_data.arguments.path_param_0
- 94: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_38) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_38: log_data.arguments.path_param_3
_api_v1_foodservice_foods@@_api_v1_orderservice_order_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_41)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_41: log_data.env.is_user
- 92: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_35) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_35: log_data.arguments.path_param_0
- 95: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_38) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_38: log_data.arguments.path_param_3
_api_v1_foodservice_foods@@_api_v1_travelservice_trips_left
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_32)
  - webnorm_invariant_gen_var_23: extra@@log_data.response.status
  - webnorm_invariant_gen_var_32: log_data.env.is_user
- 85: webnorm_invariant_gen_var_24.compareTo(webnorm_invariant_gen_var_26) < 0
  - webnorm_invariant_gen_var_24: extra@@log_data.response_time
  - webnorm_invariant_gen_var_26: log_data.arguments.path_param_0
- 88: webnorm_invariant_gen_var_24.compareTo(webnorm_invariant_gen_var_29) < 0
  - webnorm_invariant_gen_var_24: extra@@log_data.response_time
  - webnorm_invariant_gen_var_29: log_data.arguments.path_param_3
- 33: webnorm_invariant_gen_var_1.compareTo(webnorm_invariant_gen_var_29) < 0
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.departureTime
  - webnorm_invariant_gen_var_29: log_data.arguments.path_param_3

Appendix E8: Attack 8 Evaluation output

```
# Attack 8
## _api_v1_orderOtherService_orderOther_refresh
- 3: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_12)
   - webnorm_invariant_gen_var_6: log_data.arguments.loginId
   - webnorm_invariant_gen_var_12: log_data.env.user_id
##
_api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderOtherService_orderOther_re
fresh
- 9: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_46: log_data.env.user_id
## _api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderservice_order_refresh
- 9: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_45)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_45: log_data.env.is_user
## _api_v1_orderservice_order_refresh
- 3: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_12)
   - webnorm_invariant_gen_var_6: log_data.arguments.loginId
   - webnorm_invariant_gen_var_12: log_data.env.user_id
## _api_v1_orderservice_order_refresh@@_api_v1_orderservice_order_refresh
- 9: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_46: log_data.env.user_id
```

Appendix E9: Attack 9 Evaluation output

```
# Attack 9
## _api_v1_orderOtherService_orderOther_refresh
- 3: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_12)
   - webnorm_invariant_gen_var_6: log_data.arguments.loginId
   - webnorm_invariant_gen_var_12: log_data.env.user_id
##
_api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderOtherService_orderOther_re
fresh
- 9: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_46: log_data.env.user_id
## _api_v1_orderservice_order_refresh
- 3: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_12)
   - webnorm_invariant_gen_var_6: log_data.arguments.loginId
   - webnorm_invariant_gen_var_12: log_data.env.user_id
## _api_v1_orderservice_order_refresh@@_api_v1_orderservice_order_refresh
- 9: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_46: log_data.env.user_id
```

Appendix E10: Attack 10 Evaluation output

```
# Attack 10
## _api_v1_orderOtherService_orderOther_refresh
- 3: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_12)
   - webnorm_invariant_gen_var_6: log_data.arguments.loginId
   - webnorm_invariant_gen_var_12: log_data.env.user_id
##
_api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderOtherService_orderOther_re
fresh
- 9: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_46: log_data.env.user_id
## _api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderservice_order_refresh
- 9: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_45)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_45: log_data.env.is_user
## _api_v1_orderservice_order_refresh
- 3: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_12)
   - webnorm_invariant_gen_var_6: log_data.arguments.loginId
   - webnorm_invariant_gen_var_12: log_data.env.user_id
## _api_v1_orderservice_order_refresh@@_api_v1_orderservice_order_refresh
- 9: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_46: log_data.env.user_id
```

Appendix E11: Attack 11 Evaluation output

```
# Attack 11
## _api_v1_consignservice_consigns
- 2: webnorm_invariant_gen_var_4.equals("2024-11-28")
   - webnorm_invariant_gen_var_4: log_data.arguments.handleDate
## _api_v1_consignservice_consigns@@_api_v1_orderOtherService_orderOther_refresh
- 6: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_35)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_35: log_data.arguments.accountId
- 7: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_49)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_49: log_data.env.user_id
## _api_v1_consignservice_consigns@@_api_v1_orderservice_order_refresh
- 6: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_35)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_35: log_data.arguments.accountId
- 7: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_49)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_49: log_data.env.user_id
##
_api_v1_consignservice_consigns_order@@_api_v1_orderOtherService_orderOther_refresh
- 5: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_39)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_39: log_data.env.user_id
## _api_v1_consignservice_consigns_order@@_api_v1_orderservice_order_refresh
- 5: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_39)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_39: log_data.env.user_id
## _api_v1_orderOtherService_orderOther_refresh
- 3: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_12)
   - webnorm_invariant_gen_var_6: log_data.arguments.loginId
   - webnorm_invariant_gen_var_12: log_data.env.user_id
##
_api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderOtherService_orderOther_re
fresh
- 9: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_46: log_data.env.user_id
## _api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderservice_order_refresh
- 9: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_45)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_45: log_data.env.is_user
## _api_v1_orderservice_order_refresh
```

- 3: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_12)
  - webnorm_invariant_gen_var_6: log_data.arguments.loginId
  - webnorm_invariant_gen_var_12: log_data.env.user_id

_api_v1_orderservice_order_refresh@@_api_v1_orderservice_order_refresh

- 9: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_46: log_data.env.user_id

Appendix E12: Attack 12 Evaluation output

```
# Attack 12
## _api_v1_consignservice_consigns
- 2: webnorm_invariant_gen_var_4.equals("2024-11-28")
   - webnorm_invariant_gen_var_4: log_data.arguments.handleDate
## _api_v1_consignservice_consigns@@_api_v1_orderOtherService_orderOther_refresh
- 115: daikon.Quant.memberOf(webnorm_invariant_gen_var_41 ,
webnorm_invariant_gen_var_22 )
   - webnorm_invariant_gen_var_22: extra@@log_data.response.data[].id
   - webnorm_invariant_gen_var_41: log_data.arguments.orderId
## _api_v1_consignservice_consigns@@_api_v1_orderservice_order_refresh
- 117: daikon.Quant.memberOf(webnorm_invariant_gen_var_41 ,
webnorm_invariant_gen_var_22 )
   - webnorm_invariant_gen_var_22: extra@@log_data.response.data[].id
   - webnorm_invariant_gen_var_41: log_data.arguments.orderId
## _api_v1_consignservice_consigns@@_api_v1_verifycode_generate
- 14: webnorm_invariant_gen_var_2.compareTo(webnorm_invariant_gen_var_12) > 0
   - webnorm_invariant_gen_var_2: extra@@log_data.env.is_admin
   - webnorm_invariant_gen_var_12: log_data.arguments.id
## _api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderservice_order_refresh
- 9: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_45)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_45: log_data.env.is_user
```

Appendix E13: Attack 13 Evaluation output

```
# Attack 13
## _api_v1_consignservice_consigns
- 2: webnorm_invariant_gen_var_4.equals("2024-11-28")
   - webnorm_invariant_gen_var_4: log_data.arguments.handleDate
## _api_v1_consignservice_consigns@@_api_v1_orderOtherService_orderOther_refresh
- 112: daikon.Quant.memberOf(webnorm_invariant_gen_var_37 ,
webnorm_invariant_gen_var_21 )
   - webnorm_invariant_gen_var_21: extra@@log_data.response.data[].from
   - webnorm_invariant_gen_var_37: log_data.arguments.from
## _api_v1_consignservice_consigns@@_api_v1_orderservice_order_refresh
- 114: daikon.Quant.memberOf(webnorm_invariant_gen_var_37 ,
webnorm_invariant_gen_var_21 )
   - webnorm_invariant_gen_var_21: extra@@log_data.response.data[].from
   - webnorm_invariant_gen_var_37: log_data.arguments.from
## _api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderservice_order_refresh
- 9: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_45)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_45: log_data.env.is_user
```

Appendix E14: Attack 14 Evaluation output

```
# Attack 14
## _api_v1_inside_pay_service_inside_payment@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_37)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_37: log_data.env.is_user
- 58: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_34) > 0
   - webnorm_invariant_gen_var_3: extra@@log_data.arguments.from
   - webnorm_invariant_gen_var_34: log_data.arguments.tripId
- 121: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_34) > 0
   - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
   - webnorm_invariant_gen_var_34: log_data.arguments.tripId
## _api_v1_inside_pay_service_inside_payment@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_15)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_15: log_data.env.is_user
## _api_v1_inside_pay_service_inside_payment@@_api_v1_foodservice_foods
- 49: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_33) > 0
   - webnorm_invariant_gen_var_3: extra@@log_data.arguments.path_param_2
   - webnorm_invariant_gen_var_33: log_data.arguments.tripId
- 42: webnorm_invariant_gen_var_2.compareTo(webnorm_invariant_gen_var_33) > 0
   - webnorm_invariant_gen_var_2: extra@@log_data.arguments.path_param_1
   - webnorm_invariant_gen_var_33: log_data.arguments.tripId
## _api_v1_inside_pay_service_inside_payment@@_api_v1_travel2service_trips_left
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_30)
   - webnorm_invariant_gen_var_23: extra@@log_data.response.status
   - webnorm_invariant_gen_var_30: log_data.env.is_user
- 39: webnorm_invariant_gen_var_2.compareTo(webnorm_invariant_gen_var_27) > 0
   - webnorm_invariant_gen_var_2: extra@@log_data.arguments.endPlace
   - webnorm_invariant_gen_var_27: log_data.arguments.tripId
- 45: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_27) > 0
   - webnorm_invariant_gen_var_3: extra@@log_data.arguments.startPlace
   - webnorm_invariant_gen_var_27: log_data.arguments.tripId
## _api_v1_inside_pay_service_inside_payment@@_api_v1_travelservice_trips_left
- 40: webnorm_invariant_gen_var_2.compareTo(webnorm_invariant_gen_var_27) > 0
   - webnorm_invariant_gen_var_2: extra@@log_data.arguments.endPlace
   - webnorm_invariant_gen_var_27: log_data.arguments.tripId
- 46: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_27) > 0
   - webnorm_invariant_gen_var_3: extra@@log_data.arguments.startPlace
   - webnorm_invariant_gen_var_27: log_data.arguments.tripId
## _api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderservice_order_refresh
- 9: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_45)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
```

- webnorm_invariant_gen_var_45: log_data.env.is_user

Appendix E15: Attack 15 Evaluation output

```
# Attack 15
## _api_v1_inside_pay_service_inside_payment@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_37)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_37: log_data.env.is_user
- 71: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
   - webnorm_invariant_gen_var_34: log_data.arguments.tripId
- 104: webnorm_invariant_gen_var_8.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
   - webnorm_invariant_gen_var_34: log_data.arguments.tripId
- 113: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
   - webnorm_invariant_gen_var_34: log_data.arguments.tripId
- 128: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
   - webnorm_invariant_gen_var_34: log_data.arguments.tripId
- 149: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_31: extra@@log_data.response_time
   - webnorm_invariant_gen_var_34: log_data.arguments.tripId
## _api_v1_inside_pay_service_inside_payment@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_15)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_15: log_data.env.is_user
- 15: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_12) < 0
   - webnorm_invariant_gen_var_9: extra@@log_data.response_time
   - webnorm_invariant_gen_var_12: log_data.arguments.tripId
##
_api_v1_inside_pay_service_inside_payment@@_api_v1_orderOtherService_orderOther_refre
sh
- 5: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_40)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_40: log_data.env.user_id
## _api_v1_inside_pay_service_inside_payment@@_api_v1_orderservice_order_refresh
- 5: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_40)
   - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
   - webnorm_invariant_gen_var_40: log_data.env.user_id
## _api_v1_inside_pay_service_inside_payment@@_api_v1_travel2service_trips_left
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_30)
   - webnorm_invariant_gen_var_23: extra@@log_data.response.status
   - webnorm_invariant_gen_var_30: log_data.env.is_user
- 81: webnorm_invariant_gen_var_24.compareTo(webnorm_invariant_gen_var_27) < 0
```

- webnorm_invariant_gen_var_24: extra@@log_data.response_time
- webnorm_invariant_gen_var_27: log_data.arguments.tripId

- 32: webnorm_invariant_gen_var_1.compareTo(webnorm_invariant_gen_var_27) < 0
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.departureTime
  - webnorm_invariant_gen_var_27: log_data.arguments.tripId

_api_v1_orderOtherService_orderOther_refresh

- 3: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_12)
  - webnorm_invariant_gen_var_6: log_data.arguments.loginId
  - webnorm_invariant_gen_var_12: log_data.env.user_id

_api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderOtherService_orderOther_refresh

- 9: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_46: log_data.env.user_id

_api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderservice_order_refresh

- 9: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_45)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_45: log_data.env.is_user

_api_v1_orderservice_order_refresh

- 3: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_12)
  - webnorm_invariant_gen_var_6: log_data.arguments.loginId
  - webnorm_invariant_gen_var_12: log_data.env.user_id

_api_v1_orderservice_order_refresh@@_api_v1_orderservice_order_refresh

- 9: webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_46: log_data.env.user_id

Appendix E16: Attack 16 Evaluation output

```
# Attack 16
## _api_v1_inside_pay_service_inside_payment@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_37)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_37: log_data.env.is_user
- 71: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
   - webnorm_invariant_gen_var_34: log_data.arguments.tripId
- 104: webnorm_invariant_gen_var_8.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
   - webnorm_invariant_gen_var_34: log_data.arguments.tripId
- 113: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
   - webnorm_invariant_gen_var_34: log_data.arguments.tripId
- 128: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
   - webnorm_invariant_gen_var_34: log_data.arguments.tripId
- 149: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_31: extra@@log_data.response_time
   - webnorm_invariant_gen_var_34: log_data.arguments.tripId
## _api_v1_inside_pay_service_inside_payment@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_15)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_15: log_data.env.is_user
- 15: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_12) < 0
   - webnorm_invariant_gen_var_9: extra@@log_data.response_time
   - webnorm_invariant_gen_var_12: log_data.arguments.tripId
## _api_v1_inside_pay_service_inside_payment@@_api_v1_travel2service_trips_left
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_30)
   - webnorm_invariant_gen_var_23: extra@@log_data.response.status
   - webnorm_invariant_gen_var_30: log_data.env.is_user
- 81: webnorm_invariant_gen_var_24.compareTo(webnorm_invariant_gen_var_27) < 0
   - webnorm_invariant_gen_var_24: extra@@log_data.response_time
   - webnorm_invariant_gen_var_27: log_data.arguments.tripId
- 32: webnorm_invariant_gen_var_1.compareTo(webnorm_invariant_gen_var_27) < 0
   - webnorm_invariant_gen_var_1: extra@@log_data.arguments.departureTime
   - webnorm_invariant_gen_var_27: log_data.arguments.tripId
## _api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderservice_order_refresh
- 9: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_45)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_45: log_data.env.is_user
```

Appendix E17: Attack 17 Evaluation output

```
# Attack 17
## _api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderservice_order_refresh
- 9: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_45)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_45: log_data.env.is_user
## _api_v1_rebookservice_rebook@@_api_v1_assuranceservice_assurances_types
- 8: webnorm_invariant_gen_var_6.compareTo(webnorm_invariant_gen_var_9) < 0
   - webnorm_invariant_gen_var_6: extra@@log_data.response_time
   - webnorm_invariant_gen_var_9: log_data.arguments.oldTripId
## _api_v1_rebookservice_rebook@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_40)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_40: log_data.env.is_user
- 63: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_37) > 0
   - webnorm_invariant_gen_var_3: extra@@log_data.arguments.from
   - webnorm_invariant_gen_var_37: log_data.arguments.tripId
- 75: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_33) <= 0
   - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
   - webnorm_invariant_gen_var_33: log_data.arguments.date
- 76: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
   - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 115: webnorm_invariant_gen_var_8.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
   - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 126: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
   - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 138: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_37) > 0
   - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
   - webnorm_invariant_gen_var_37: log_data.arguments.tripId
- 145: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
   - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 174: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_31: extra@@log_data.response_time
   - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
## _api_v1_rebookservice_rebook@@_api_v1_consignservice_consigns_account
- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_30)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_30: log_data.env.is_user
- 88: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_24) < 0
```

- webnorm_invariant_gen_var_21: extra@@log_data.response_time
- webnorm_invariant_gen_var_24: log_data.arguments.oldTripId

_api_v1_rebookservice_rebook@@_api_v1_consignservice_consigns_order

- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_18)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_18: log_data.env.is_user
- 19: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_12) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_12: log_data.arguments.oldTripId

_api_v1_rebookservice_rebook@@_api_v1_contactservice_contacts_account

- 34: webnorm_invariant_gen_var_16.compareTo(webnorm_invariant_gen_var_19) < 0
  - webnorm_invariant_gen_var_16: extra@@log_data.response_time
  - webnorm_invariant_gen_var_19: log_data.arguments.oldTripId
- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_25)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_25: log_data.env.is_user

_api_v1_rebookservice_rebook@@_api_v1_foodservice_foods

- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_7, webnorm_invariant_gen_var_39)
  - webnorm_invariant_gen_var_7: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_39: log_data.env.is_user
- 32: webnorm_invariant_gen_var_1.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_33: log_data.arguments.oldTripId
- 101: webnorm_invariant_gen_var_30.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_30: extra@@log_data.response_time
  - webnorm_invariant_gen_var_33: log_data.arguments.oldTripId
- 44: webnorm_invariant_gen_var_2.compareTo(webnorm_invariant_gen_var_36) > 0
  - webnorm_invariant_gen_var_2: extra@@log_data.arguments.path_param_1
  - webnorm_invariant_gen_var_36: log_data.arguments.tripId
- 53: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_36) > 0
  - webnorm_invariant_gen_var_3: extra@@log_data.arguments.path_param_2
  - webnorm_invariant_gen_var_36: log_data.arguments.tripId

_api_v1_rebookservice_rebook@@_api_v1_orderOtherService_orderOther_refresh

- 105: daikon.Quant.memberOf(webnorm_invariant_gen_var_37 , webnorm_invariant_gen_var_22)
  - webnorm_invariant_gen_var_22: extra@@log_data.response.data[].id
  - webnorm_invariant_gen_var_37: log_data.arguments.orderId
- 133: daikon.Quant.memberOf(webnorm_invariant_gen_var_36 , webnorm_invariant_gen_var_28)
  - webnorm_invariant_gen_var_28: extra@@log_data.response.data[].trainNumber
  - webnorm_invariant_gen_var_36: log_data.arguments.oldTripId

_api_v1_rebookservice_rebook@@_api_v1_orderservice_order_refresh

- 107: daikon.Quant.memberOf(webnorm_invariant_gen_var_37 , webnorm_invariant_gen_var_22)

- webnorm_invariant_gen_var_22: extra@@log_data.response.data[].id
- webnorm_invariant_gen_var_37: log_data.arguments.orderId

- 135: daikon.Quant.memberOf(webnorm_invariant_gen_var_36 , webnorm_invariant_gen_var_28)
  - webnorm_invariant_gen_var_28: extra@@log_data.response.data[].trainNumber
  - webnorm_invariant_gen_var_36: log_data.arguments.oldTripId

_api_v1_rebookservice_rebook@@_api_v1_travelservice_trips_left

- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_33)
  - webnorm_invariant_gen_var_23: extra@@log_data.response.status
  - webnorm_invariant_gen_var_33: log_data.env.is_user
- 31: webnorm_invariant_gen_var_1.compareTo(webnorm_invariant_gen_var_27) < 0
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.departureTime
  - webnorm_invariant_gen_var_27: log_data.arguments.oldTripId
- 90: webnorm_invariant_gen_var_24.compareTo(webnorm_invariant_gen_var_27) < 0
  - webnorm_invariant_gen_var_24: extra@@log_data.response_time
  - webnorm_invariant_gen_var_27: log_data.arguments.oldTripId
- 41: webnorm_invariant_gen_var_2.compareTo(webnorm_invariant_gen_var_28) > 0
  - webnorm_invariant_gen_var_2: extra@@log_data.arguments.endPlace
  - webnorm_invariant_gen_var_28: log_data.arguments.orderId
- 42: webnorm_invariant_gen_var_2.compareTo(webnorm_invariant_gen_var_30) > 0
  - webnorm_invariant_gen_var_2: extra@@log_data.arguments.endPlace
  - webnorm_invariant_gen_var_30: log_data.arguments.tripId
- 50: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_30) > 0
  - webnorm_invariant_gen_var_3: extra@@log_data.arguments.startPlace
  - webnorm_invariant_gen_var_30: log_data.arguments.tripId

Appendix E18: Attack 18 Evaluation output

```
# Attack 18
## _api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderservice_order_refresh
- 9: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_45)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_45: log_data.env.is_user
## _api_v1_rebookservice_rebook@@_api_v1_assuranceservice_assurances_types
- 8: webnorm_invariant_gen_var_6.compareTo(webnorm_invariant_gen_var_9) < 0
   - webnorm_invariant_gen_var_6: extra@@log_data.response_time
   - webnorm_invariant_gen_var_9: log_data.arguments.oldTripId
- 10: webnorm_invariant_gen_var_6.compareTo(webnorm_invariant_gen_var_12) < 0
   - webnorm_invariant_gen_var_6: extra@@log_data.response_time
   - webnorm_invariant_gen_var_12: log_data.arguments.tripId
## _api_v1_rebookservice_rebook@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_40)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_40: log_data.env.is_user
- 75: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_33) <= 0
   - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
   - webnorm_invariant_gen_var_33: log_data.arguments.date
- 76: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
   - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 78: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_37) < 0
   - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
   - webnorm_invariant_gen_var_37: log_data.arguments.tripId
- 115: webnorm_invariant_gen_var_8.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
   - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 117: webnorm_invariant_gen_var_8.compareTo(webnorm_invariant_gen_var_37) < 0
   - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
   - webnorm_invariant_gen_var_37: log_data.arguments.tripId
- 126: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
   - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 128: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_37) < 0
   - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
   - webnorm_invariant_gen_var_37: log_data.arguments.tripId
- 145: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
   - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 147: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_37) < 0
   - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
```

  - webnorm_invariant_gen_var_37: log_data.arguments.tripId
- 174: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_34) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 176: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_37) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_37: log_data.arguments.tripId
_api_v1_rebookservice_rebook@@_api_v1_consignservice_consigns_account
- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_30)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_30: log_data.env.is_user
- 88: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_24) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_24: log_data.arguments.oldTripId
- 90: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_27) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_27: log_data.arguments.tripId
_api_v1_rebookservice_rebook@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_18)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_18: log_data.env.is_user
- 19: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_12) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_12: log_data.arguments.oldTripId
- 21: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_15) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_15: log_data.arguments.tripId
_api_v1_rebookservice_rebook@@_api_v1_contactservice_contacts_account
- 34: webnorm_invariant_gen_var_16.compareTo(webnorm_invariant_gen_var_19) < 0
  - webnorm_invariant_gen_var_16: extra@@log_data.response_time
  - webnorm_invariant_gen_var_19: log_data.arguments.oldTripId
- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_25)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_25: log_data.env.is_user
- 36: webnorm_invariant_gen_var_16.compareTo(webnorm_invariant_gen_var_22) < 0
  - webnorm_invariant_gen_var_16: extra@@log_data.response_time
  - webnorm_invariant_gen_var_22: log_data.arguments.tripId
_api_v1_rebookservice_rebook@@_api_v1_foodservice_foods
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_7, webnorm_invariant_gen_var_39)
  - webnorm_invariant_gen_var_7: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_39: log_data.env.is_user
- 32: webnorm_invariant_gen_var_1.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_33: log_data.arguments.oldTripId

- 34: webnorm_invariant_gen_var_1.compareTo(webnorm_invariant_gen_var_36) < 0
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_36: log_data.arguments.tripId
- 101: webnorm_invariant_gen_var_30.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_30: extra@@log_data.response_time
  - webnorm_invariant_gen_var_33: log_data.arguments.oldTripId
- 103: webnorm_invariant_gen_var_30.compareTo(webnorm_invariant_gen_var_36) < 0
  - webnorm_invariant_gen_var_30: extra@@log_data.response_time
  - webnorm_invariant_gen_var_36: log_data.arguments.tripId

Appendix E19: Attack 19 Evaluation output

```
# Attack 19
## _api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderservice_order_refresh
- 9: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_45)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_45: log_data.env.is_user
## _api_v1_rebookservice_rebook@@_api_v1_assuranceservice_assurances_types
- 8: webnorm_invariant_gen_var_6.compareTo(webnorm_invariant_gen_var_9) < 0
   - webnorm_invariant_gen_var_6: extra@@log_data.response_time
   - webnorm_invariant_gen_var_9: log_data.arguments.oldTripId
- 10: webnorm_invariant_gen_var_6.compareTo(webnorm_invariant_gen_var_12) < 0
   - webnorm_invariant_gen_var_6: extra@@log_data.response_time
   - webnorm_invariant_gen_var_12: log_data.arguments.tripId
## _api_v1_rebookservice_rebook@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_40)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_40: log_data.env.is_user
- 75: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_33) <= 0
   - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
   - webnorm_invariant_gen_var_33: log_data.arguments.date
- 76: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
   - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 78: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_37) < 0
   - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
   - webnorm_invariant_gen_var_37: log_data.arguments.tripId
- 115: webnorm_invariant_gen_var_8.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
   - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 117: webnorm_invariant_gen_var_8.compareTo(webnorm_invariant_gen_var_37) < 0
   - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
   - webnorm_invariant_gen_var_37: log_data.arguments.tripId
- 126: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
   - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 128: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_37) < 0
   - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
   - webnorm_invariant_gen_var_37: log_data.arguments.tripId
- 145: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
   - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 147: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_37) < 0
   - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
```

  - webnorm_invariant_gen_var_37: log_data.arguments.tripId
- 174: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_34) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 176: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_37) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_37: log_data.arguments.tripId
_api_v1_rebookservice_rebook@@_api_v1_consignservice_consigns_account
- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_30)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_30: log_data.env.is_user
- 88: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_24) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_24: log_data.arguments.oldTripId
- 90: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_27) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_27: log_data.arguments.tripId
_api_v1_rebookservice_rebook@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_18)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_18: log_data.env.is_user
- 19: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_12) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_12: log_data.arguments.oldTripId
- 21: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_15) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_15: log_data.arguments.tripId
_api_v1_rebookservice_rebook@@_api_v1_contactservice_contacts_account
- 34: webnorm_invariant_gen_var_16.compareTo(webnorm_invariant_gen_var_19) < 0
  - webnorm_invariant_gen_var_16: extra@@log_data.response_time
  - webnorm_invariant_gen_var_19: log_data.arguments.oldTripId
- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_25)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_25: log_data.env.is_user
- 36: webnorm_invariant_gen_var_16.compareTo(webnorm_invariant_gen_var_22) < 0
  - webnorm_invariant_gen_var_16: extra@@log_data.response_time
  - webnorm_invariant_gen_var_22: log_data.arguments.tripId
_api_v1_rebookservice_rebook@@_api_v1_foodservice_foods
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_7, webnorm_invariant_gen_var_39)
  - webnorm_invariant_gen_var_7: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_39: log_data.env.is_user
- 32: webnorm_invariant_gen_var_1.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_33: log_data.arguments.oldTripId

- 34: webnorm_invariant_gen_var_1.compareTo(webnorm_invariant_gen_var_36) < 0
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_36: log_data.arguments.tripId
- 101: webnorm_invariant_gen_var_30.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_30: extra@@log_data.response_time
  - webnorm_invariant_gen_var_33: log_data.arguments.oldTripId
- 103: webnorm_invariant_gen_var_30.compareTo(webnorm_invariant_gen_var_36) < 0
  - webnorm_invariant_gen_var_30: extra@@log_data.response_time
  - webnorm_invariant_gen_var_36: log_data.arguments.tripId

Appendix E20: Attack 20 Evaluation output

```
# Attack 20
## _api_v1_orderOtherService_orderOther_refresh@@_api_v1_orderservice_order_refresh
- 9: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_45)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_45: log_data.env.is_user
## _api_v1_rebookservice_rebook@@_api_v1_assuranceservice_assurances_types
- 8: webnorm_invariant_gen_var_6.compareTo(webnorm_invariant_gen_var_9) < 0
   - webnorm_invariant_gen_var_6: extra@@log_data.response_time
   - webnorm_invariant_gen_var_9: log_data.arguments.oldTripId
- 10: webnorm_invariant_gen_var_6.compareTo(webnorm_invariant_gen_var_12) < 0
   - webnorm_invariant_gen_var_6: extra@@log_data.response_time
   - webnorm_invariant_gen_var_12: log_data.arguments.tripId
## _api_v1_rebookservice_rebook@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_40)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_40: log_data.env.is_user
- 75: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_33) <= 0
   - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
   - webnorm_invariant_gen_var_33: log_data.arguments.date
- 76: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
   - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 78: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_37) < 0
   - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
   - webnorm_invariant_gen_var_37: log_data.arguments.tripId
- 115: webnorm_invariant_gen_var_8.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
   - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 117: webnorm_invariant_gen_var_8.compareTo(webnorm_invariant_gen_var_37) < 0
   - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
   - webnorm_invariant_gen_var_37: log_data.arguments.tripId
- 126: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
   - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 128: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_37) < 0
   - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
   - webnorm_invariant_gen_var_37: log_data.arguments.tripId
- 139: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_41) > 0
   - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
   - webnorm_invariant_gen_var_41: log_data.env.user_id
- 145: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_34) < 0
   - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
```

  - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 147: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_37) < 0
  - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
  - webnorm_invariant_gen_var_37: log_data.arguments.tripId
- 174: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_34) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_34: log_data.arguments.oldTripId
- 176: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_37) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_37: log_data.arguments.tripId
_api_v1_rebookservice_rebook@@_api_v1_consignservice_consigns_account
- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_30)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_30: log_data.env.is_user
- 88: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_24) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_24: log_data.arguments.oldTripId
- 90: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_27) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_27: log_data.arguments.tripId
_api_v1_rebookservice_rebook@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_18)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_18: log_data.env.is_user
- 19: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_12) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_12: log_data.arguments.oldTripId
- 21: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_15) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_15: log_data.arguments.tripId
_api_v1_rebookservice_rebook@@_api_v1_contactservice_contacts_account
- 34: webnorm_invariant_gen_var_16.compareTo(webnorm_invariant_gen_var_19) < 0
  - webnorm_invariant_gen_var_16: extra@@log_data.response_time
  - webnorm_invariant_gen_var_19: log_data.arguments.oldTripId
- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_25)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_25: log_data.env.is_user
- 36: webnorm_invariant_gen_var_16.compareTo(webnorm_invariant_gen_var_22) < 0
  - webnorm_invariant_gen_var_16: extra@@log_data.response_time
  - webnorm_invariant_gen_var_22: log_data.arguments.tripId
_api_v1_rebookservice_rebook@@_api_v1_foodservice_foods
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_7, webnorm_invariant_gen_var_39)
  - webnorm_invariant_gen_var_7: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_39: log_data.env.is_user

- 32: webnorm_invariant_gen_var_1.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_33: log_data.arguments.oldTripId
- 34: webnorm_invariant_gen_var_1.compareTo(webnorm_invariant_gen_var_36) < 0
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_36: log_data.arguments.tripId
- 101: webnorm_invariant_gen_var_30.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_30: extra@@log_data.response_time
  - webnorm_invariant_gen_var_33: log_data.arguments.oldTripId
- 103: webnorm_invariant_gen_var_30.compareTo(webnorm_invariant_gen_var_36) < 0
  - webnorm_invariant_gen_var_30: extra@@log_data.response_time
  - webnorm_invariant_gen_var_36: log_data.arguments.tripId
- 54: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_40) > 0
  - webnorm_invariant_gen_var_3: extra@@log_data.arguments.path_param_2
  - webnorm_invariant_gen_var_40: log_data.env.user_id

_api_v1_rebookservice_rebook@@_api_v1_orderOtherService_orderOther_refresh

- 133: daikon.Quant.memberOf(webnorm_invariant_gen_var_36 , webnorm_invariant_gen_var_28)
  - webnorm_invariant_gen_var_28: extra@@log_data.response.data[].trainNumber
  - webnorm_invariant_gen_var_36: log_data.arguments.oldTripId

_api_v1_rebookservice_rebook@@_api_v1_orderservice_order_refresh

- 135: daikon.Quant.memberOf(webnorm_invariant_gen_var_36 , webnorm_invariant_gen_var_28)
  - webnorm_invariant_gen_var_28: extra@@log_data.response.data[].trainNumber
  - webnorm_invariant_gen_var_36: log_data.arguments.oldTripId

_api_v1_rebookservice_rebook@@_api_v1_travel2service_trips_left

- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_33)
  - webnorm_invariant_gen_var_23: extra@@log_data.response.status
  - webnorm_invariant_gen_var_33: log_data.env.is_user
- 31: webnorm_invariant_gen_var_1.compareTo(webnorm_invariant_gen_var_27) < 0
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.departureTime
  - webnorm_invariant_gen_var_27: log_data.arguments.oldTripId
- 90: webnorm_invariant_gen_var_24.compareTo(webnorm_invariant_gen_var_27) < 0
  - webnorm_invariant_gen_var_24: extra@@log_data.response_time
  - webnorm_invariant_gen_var_27: log_data.arguments.oldTripId
- 33: webnorm_invariant_gen_var_1.compareTo(webnorm_invariant_gen_var_30) < 0
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.departureTime
  - webnorm_invariant_gen_var_30: log_data.arguments.tripId
- 92: webnorm_invariant_gen_var_24.compareTo(webnorm_invariant_gen_var_30) < 0
  - webnorm_invariant_gen_var_24: extra@@log_data.response_time
  - webnorm_invariant_gen_var_30: log_data.arguments.tripId

_api_v1_travelservice_trips_left@@_api_v1_consignservice_consigns

- 125: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_39) > 0
  - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to

- webnorm_invariant_gen_var_39: log_data.env.user_id

Appendix E21: Attack 21 Evaluation output

```
# Attack 21
## _api_v1_contactservice_contacts_account
- 0: webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_5)
   - webnorm_invariant_gen_var_1: log_data.arguments.path_param_0
   - webnorm_invariant_gen_var_5: log_data.env.user_id
##
_api_v1_contactservice_contacts_account@@_api_v1_assuranceservice_assurances_types
- 4: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_8) >= 0
   - webnorm_invariant_gen_var_4: extra@@log_data.env.user_id
   - webnorm_invariant_gen_var_8: log_data.arguments.path_param_0
## _api_v1_contactservice_contacts_account@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_36)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_36: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_consignservice_consigns_account
- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_26)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_26: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_14)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_14: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_contactservice_contacts_account
- 2: webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_22)
   - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
   - webnorm_invariant_gen_var_22: log_data.env.user_id
##
_api_v1_contactservice_contacts_account@@_api_v1_orderOtherService_orderOther_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_38)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_38: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_orderservice_order_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_38)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_38: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_users_login
- 2: webnorm_invariant_gen_var_11.equals(webnorm_invariant_gen_var_17)
   - webnorm_invariant_gen_var_11: extra@@log_data.response.data.userId
   - webnorm_invariant_gen_var_17: log_data.arguments.path_param_0
## _api_v1_foodservice_foods@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_39)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
```

  - webnorm_invariant_gen_var_39: log_data.env.is_user
- 61: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_36) > 0
  - webnorm_invariant_gen_var_3: extra@@log_data.arguments.from
  - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
- 73: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
  - webnorm_invariant_gen_var_33: log_data.arguments.path_param_0
- 124: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
  - webnorm_invariant_gen_var_33: log_data.arguments.path_param_0
- 135: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_36) > 0
  - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
  - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
- 171: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_33: log_data.arguments.path_param_0
_api_v1_foodservice_foods@@_api_v1_consignservice_consigns_account
- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_29)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_29: log_data.env.is_user
- 66: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_23) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_23: log_data.arguments.path_param_0
_api_v1_foodservice_foods@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_17)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_17: log_data.env.is_user
- 19: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_11) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_11: log_data.arguments.path_param_0
_api_v1_foodservice_foods@@_api_v1_orderOtherService_orderOther_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_41)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_41: log_data.env.is_user
- 91: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_35) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_35: log_data.arguments.path_param_0
_api_v1_foodservice_foods@@_api_v1_orderservice_order_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_41)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_41: log_data.env.is_user
- 92: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_35) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_35: log_data.arguments.path_param_0

_api_v1_preserveotherservice_preserveOther@@_api_v1_consignservice_consigns
- 13: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_42)
  - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_42: log_data.arguments.foodType
- 14: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_53)
  - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_53: log_data.env.is_user
- 41: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.accountId
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 42: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.accountId
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 47: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.accountId
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 66: !webnorm_invariant_gen_var_2.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_2: extra@@log_data.arguments.consignee
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 90: !webnorm_invariant_gen_var_3.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_3: extra@@log_data.arguments.from
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 91: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_36) > 0
  - webnorm_invariant_gen_var_3: extra@@log_data.arguments.from
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 95: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_44) > 0
  - webnorm_invariant_gen_var_3: extra@@log_data.arguments.from
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 98: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_50) > 0
  - webnorm_invariant_gen_var_3: extra@@log_data.arguments.from
  - webnorm_invariant_gen_var_50: log_data.arguments.tripId
- 111: !webnorm_invariant_gen_var_4.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 112: !webnorm_invariant_gen_var_4.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 114: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_39) < 0
  - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
  - webnorm_invariant_gen_var_39: log_data.arguments.date
- 115: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName
- 132: !webnorm_invariant_gen_var_5.equals(webnorm_invariant_gen_var_35)

  - webnorm_invariant_gen_var_5: extra@@log_data.arguments.id
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 133: !webnorm_invariant_gen_var_5.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_5: extra@@log_data.arguments.id
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 138: !webnorm_invariant_gen_var_5.equals(webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_5: extra@@log_data.arguments.id
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 155: !webnorm_invariant_gen_var_7.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_7: extra@@log_data.arguments.orderId
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 156: !webnorm_invariant_gen_var_7.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_7: extra@@log_data.arguments.orderId
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 161: !webnorm_invariant_gen_var_7.equals(webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_7: extra@@log_data.arguments.orderId
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 175: !webnorm_invariant_gen_var_8.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 176: !webnorm_invariant_gen_var_8.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 179: webnorm_invariant_gen_var_8.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName
- 181: !webnorm_invariant_gen_var_8.equals(webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 194: !webnorm_invariant_gen_var_9.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 195: !webnorm_invariant_gen_var_9.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 197: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_39) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
  - webnorm_invariant_gen_var_39: log_data.arguments.date
- 198: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName
- 200: !webnorm_invariant_gen_var_9.equals(webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate

- 212: !webnorm_invariant_gen_var_10.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 213: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_36) > 0
  - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 214: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_38) > 0
  - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
  - webnorm_invariant_gen_var_38: log_data.arguments.contactsId
- 217: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_44) > 0
  - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 220: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_50) > 0
  - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
  - webnorm_invariant_gen_var_50: log_data.arguments.tripId
- 227: !webnorm_invariant_gen_var_11.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 228: !webnorm_invariant_gen_var_11.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 231: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName
- 233: !webnorm_invariant_gen_var_11.equals(webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 245: !webnorm_invariant_gen_var_22.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_22: extra@@log_data.response.data.id
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 246: !webnorm_invariant_gen_var_22.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_22: extra@@log_data.response.data.id
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 251: !webnorm_invariant_gen_var_22.equals(webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_22: extra@@log_data.response.data.id
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 260: !webnorm_invariant_gen_var_23.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_23: extra@@log_data.response.data.orderId
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 261: !webnorm_invariant_gen_var_23.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_23: extra@@log_data.response.data.orderId
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 266: !webnorm_invariant_gen_var_23.equals(webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_23: extra@@log_data.response.data.orderId

  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 273: daikon.Quant.fuzzy.gt(webnorm_invariant_gen_var_25, webnorm_invariant_gen_var_37)
  - webnorm_invariant_gen_var_25: extra@@log_data.response.data.price
  - webnorm_invariant_gen_var_37: log_data.arguments.consigneeWeight
- 274: daikon.Quant.fuzzy.gt(webnorm_invariant_gen_var_25, webnorm_invariant_gen_var_41)
  - webnorm_invariant_gen_var_25: extra@@log_data.response.data.price
  - webnorm_invariant_gen_var_41: log_data.arguments.foodPrice
- 275: daikon.Quant.fuzzy.gt(webnorm_invariant_gen_var_28, webnorm_invariant_gen_var_41)
  - webnorm_invariant_gen_var_28: extra@@log_data.response.data.weight
  - webnorm_invariant_gen_var_41: log_data.arguments.foodPrice
- 279: !webnorm_invariant_gen_var_29.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_29: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 280: webnorm_invariant_gen_var_29.compareTo(webnorm_invariant_gen_var_36) > 0
  - webnorm_invariant_gen_var_29: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 285: webnorm_invariant_gen_var_29.compareTo(webnorm_invariant_gen_var_44) > 0
  - webnorm_invariant_gen_var_29: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 292: !webnorm_invariant_gen_var_31.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 293: !webnorm_invariant_gen_var_31.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 295: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_39) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_39: log_data.arguments.date
- 296: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName

_api_v1_preserveotherservice_preserveOther@@_api_v1_consignservice_consigns_account

- 98: !webnorm_invariant_gen_var_21.equals(webnorm_invariant_gen_var_25)
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_25: log_data.arguments.consigneeName
- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_32)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_32: log_data.arguments.foodType
- 4: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_43)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_43: log_data.env.is_user
- 99: !webnorm_invariant_gen_var_21.equals(webnorm_invariant_gen_var_26)
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time

  - webnorm_invariant_gen_var_26: log_data.arguments.consigneePhone
- 101: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_29) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_29: log_data.arguments.date
- 102: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_30) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_30: log_data.arguments.foodName
- 43: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_25)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_25: log_data.arguments.consigneeName
- 44: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_26)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_26: log_data.arguments.consigneePhone
- 49: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_34)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_34: log_data.arguments.handleDate
- 84: !webnorm_invariant_gen_var_19.equals(webnorm_invariant_gen_var_25)
  - webnorm_invariant_gen_var_19: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_25: log_data.arguments.consigneeName

_api_v1_preserveotherservice_preserveOther@@_api_v1_consignservice_consigns_order

- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_20)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_20: log_data.arguments.foodType
- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_31)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_31: log_data.env.is_user
- 36: !webnorm_invariant_gen_var_5.equals(webnorm_invariant_gen_var_13)
  - webnorm_invariant_gen_var_5: extra@@log_data.env.user_id
  - webnorm_invariant_gen_var_13: log_data.arguments.consigneeName
- 37: !webnorm_invariant_gen_var_5.equals(webnorm_invariant_gen_var_14)
  - webnorm_invariant_gen_var_5: extra@@log_data.env.user_id
  - webnorm_invariant_gen_var_14: log_data.arguments.consigneePhone
- 42: !webnorm_invariant_gen_var_5.equals(webnorm_invariant_gen_var_22)
  - webnorm_invariant_gen_var_5: extra@@log_data.env.user_id
  - webnorm_invariant_gen_var_22: log_data.arguments.handleDate
- 49: !webnorm_invariant_gen_var_9.equals(webnorm_invariant_gen_var_13)
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_13: log_data.arguments.consigneeName
- 50: !webnorm_invariant_gen_var_9.equals(webnorm_invariant_gen_var_14)
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_14: log_data.arguments.consigneePhone
- 22: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_13)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_13: log_data.arguments.consigneeName

- 23: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_14)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_14: log_data.arguments.consigneePhone
- 52: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_17) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_17: log_data.arguments.date
- 53: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_18) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_18: log_data.arguments.foodName
- 28: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_22)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_22: log_data.arguments.handleDate

_api_v1_preserveotherservice_preserveOther@@_api_v1_contactservice_contacts_account
- 1: webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_18)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_18: log_data.arguments.accountId
- 2: webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_39)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_39: log_data.env.user_id

_api_v1_preserveotherservice_preserveOther@@_api_v1_inside_pay_service_inside_payment
- 39: !webnorm_invariant_gen_var_2.equals(webnorm_invariant_gen_var_15)
  - webnorm_invariant_gen_var_2: extra@@log_data.arguments.tripId
  - webnorm_invariant_gen_var_15: log_data.arguments.consigneeName
- 71: !webnorm_invariant_gen_var_9.equals(webnorm_invariant_gen_var_15)
  - webnorm_invariant_gen_var_9: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_15: log_data.arguments.consigneeName
- 72: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_16) > 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_16: log_data.arguments.consigneePhone
- 77: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_24) > 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_24: log_data.arguments.handleDate
- 84: !webnorm_invariant_gen_var_11.equals(webnorm_invariant_gen_var_15)
  - webnorm_invariant_gen_var_11: extra@@log_data.response_time
  - webnorm_invariant_gen_var_15: log_data.arguments.consigneeName
- 85: !webnorm_invariant_gen_var_11.equals(webnorm_invariant_gen_var_16)
  - webnorm_invariant_gen_var_11: extra@@log_data.response_time
  - webnorm_invariant_gen_var_16: log_data.arguments.consigneePhone
- 23: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_15)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.orderId
  - webnorm_invariant_gen_var_15: log_data.arguments.consigneeName
- 24: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_16)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.orderId

  - webnorm_invariant_gen_var_16: log_data.arguments.consigneePhone
- 57: !webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_15)
  - webnorm_invariant_gen_var_6: extra@@log_data.env.user_id
  - webnorm_invariant_gen_var_15: log_data.arguments.consigneeName
- 58: !webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_16)
  - webnorm_invariant_gen_var_6: extra@@log_data.env.user_id
  - webnorm_invariant_gen_var_16: log_data.arguments.consigneePhone
- 29: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_24)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.orderId
  - webnorm_invariant_gen_var_24: log_data.arguments.handleDate
- 63: !webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_24)
  - webnorm_invariant_gen_var_6: extra@@log_data.env.user_id
  - webnorm_invariant_gen_var_24: log_data.arguments.handleDate
_api_v1_preserveotherservice_preserveOther@@_api_v1_orderOtherService_orderOther_refresh
- 64: !webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_37)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_37: log_data.arguments.consigneeName
- 161: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_46) > 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_46: log_data.arguments.handleDate
- 65: !webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_38)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_38: log_data.arguments.consigneePhone
- 70: !webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_46: log_data.arguments.handleDate
- 7: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_44: log_data.arguments.foodType
- 8: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_55)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_55: log_data.env.is_user
- 168: !webnorm_invariant_gen_var_33.equals(webnorm_invariant_gen_var_37)
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_37: log_data.arguments.consigneeName
- 169: !webnorm_invariant_gen_var_33.equals(webnorm_invariant_gen_var_38)
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_38: log_data.arguments.consigneePhone
- 171: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_41) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_41: log_data.arguments.date
- 172: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_42) < 0

  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_42: log_data.arguments.foodName
- 155: !webnorm_invariant_gen_var_31.equals(webnorm_invariant_gen_var_37)
  - webnorm_invariant_gen_var_31: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_37: log_data.arguments.consigneeName
- 156: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_38) > 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_38: log_data.arguments.consigneePhone
_api_v1_preserveotherservice_preserveOther@@_api_v1_orderservice_order_refresh
- 64: !webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_37)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_37: log_data.arguments.consigneeName
- 65: !webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_38)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_38: log_data.arguments.consigneePhone
- 163: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_46) > 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_46: log_data.arguments.handleDate
- 70: !webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_46: log_data.arguments.handleDate
- 7: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_44: log_data.arguments.foodType
- 8: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_55)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_55: log_data.env.is_user
- 170: !webnorm_invariant_gen_var_33.equals(webnorm_invariant_gen_var_37)
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_37: log_data.arguments.consigneeName
- 171: !webnorm_invariant_gen_var_33.equals(webnorm_invariant_gen_var_38)
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_38: log_data.arguments.consigneePhone
- 173: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_41) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_41: log_data.arguments.date
- 174: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_42) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_42: log_data.arguments.foodName
- 157: !webnorm_invariant_gen_var_31.equals(webnorm_invariant_gen_var_37)
  - webnorm_invariant_gen_var_31: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_37: log_data.arguments.consigneeName
- 158: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_38) > 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response.msg

- webnorm_invariant_gen_var_38: log_data.arguments.consigneePhone

Appendix E22: Attack 22 Evaluation output

```
# Attack 22
## _api_v1_contactservice_contacts_account@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_36)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_36: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_consignservice_consigns_account
- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_26)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_26: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_14)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_14: log_data.env.is_user
##
_api_v1_contactservice_contacts_account@@_api_v1_orderOtherService_orderOther_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_38)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_38: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_orderservice_order_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_38)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_38: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_travelservice_trips_left
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_29)
   - webnorm_invariant_gen_var_23: extra@@log_data.response.status
   - webnorm_invariant_gen_var_29: log_data.env.is_user
## _api_v1_foodservice_foods@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_39)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_39: log_data.env.is_user
- 61: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_36) > 0
   - webnorm_invariant_gen_var_3: extra@@log_data.arguments.from
   - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
- 73: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_33) < 0
   - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
   - webnorm_invariant_gen_var_33: log_data.arguments.path_param_0
- 124: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_33) < 0
   - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
   - webnorm_invariant_gen_var_33: log_data.arguments.path_param_0
- 135: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_36) > 0
   - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
   - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
```

- 171: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_33: log_data.arguments.path_param_0

_api_v1_foodservice_foods@@_api_v1_consignservice_consigns_account

- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_29)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_29: log_data.env.is_user
- 66: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_23) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_23: log_data.arguments.path_param_0

_api_v1_foodservice_foods@@_api_v1_consignservice_consigns_order

- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_17)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_17: log_data.env.is_user
- 19: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_11) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_11: log_data.arguments.path_param_0

_api_v1_foodservice_foods@@_api_v1_orderOtherService_orderOther_refresh

- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_41)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_41: log_data.env.is_user
- 91: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_35) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_35: log_data.arguments.path_param_0

_api_v1_foodservice_foods@@_api_v1_orderservice_order_refresh

- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_41)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_41: log_data.env.is_user
- 92: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_35) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_35: log_data.arguments.path_param_0

_api_v1_foodservice_foods@@_api_v1_travelservice_trips_left

- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_32)
  - webnorm_invariant_gen_var_23: extra@@log_data.response.status
  - webnorm_invariant_gen_var_32: log_data.env.is_user
- 85: webnorm_invariant_gen_var_24.compareTo(webnorm_invariant_gen_var_26) < 0
  - webnorm_invariant_gen_var_24: extra@@log_data.response_time
  - webnorm_invariant_gen_var_26: log_data.arguments.path_param_0
- 40: webnorm_invariant_gen_var_2.compareTo(webnorm_invariant_gen_var_29) > 0
  - webnorm_invariant_gen_var_2: extra@@log_data.arguments.endPlace
  - webnorm_invariant_gen_var_29: log_data.arguments.path_param_3
- 46: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_29) > 0
  - webnorm_invariant_gen_var_3: extra@@log_data.arguments.startPlace
  - webnorm_invariant_gen_var_29: log_data.arguments.path_param_3

_api_v1_preserveotherservice_preserveOther
- 0: webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_22)
  - webnorm_invariant_gen_var_1: log_data.arguments.accountId
  - webnorm_invariant_gen_var_22: log_data.env.user_id
_api_v1_preserveotherservice_preserveOther@@_api_v1_assuranceservice_assurances_types
- 0: webnorm_invariant_gen_var_4.equals(webnorm_invariant_gen_var_8)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.user_id
  - webnorm_invariant_gen_var_8: log_data.arguments.accountId
_api_v1_preserveotherservice_preserveOther@@_api_v1_consignservice_consigns
- 13: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_42)
  - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_42: log_data.arguments.foodType
- 14: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_53)
  - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_53: log_data.env.is_user
- 273: daikon.Quant.fuzzy.gt(webnorm_invariant_gen_var_25, webnorm_invariant_gen_var_37)
  - webnorm_invariant_gen_var_25: extra@@log_data.response.data.price
  - webnorm_invariant_gen_var_37: log_data.arguments.consigneeWeight
- 274: daikon.Quant.fuzzy.gt(webnorm_invariant_gen_var_25, webnorm_invariant_gen_var_41)
  - webnorm_invariant_gen_var_25: extra@@log_data.response.data.price
  - webnorm_invariant_gen_var_41: log_data.arguments.foodPrice
- 275: daikon.Quant.fuzzy.gt(webnorm_invariant_gen_var_28, webnorm_invariant_gen_var_41)
  - webnorm_invariant_gen_var_28: extra@@log_data.response.data.weight
  - webnorm_invariant_gen_var_41: log_data.arguments.foodPrice
- 295: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_39) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_39: log_data.arguments.date
- 296: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName
- 98: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_50) > 0
  - webnorm_invariant_gen_var_3: extra@@log_data.arguments.from
  - webnorm_invariant_gen_var_50: log_data.arguments.tripId
- 114: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_39) < 0
  - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
  - webnorm_invariant_gen_var_39: log_data.arguments.date
- 115: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName
- 179: webnorm_invariant_gen_var_8.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName

- 197: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_39) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
  - webnorm_invariant_gen_var_39: log_data.arguments.date
- 198: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName
- 214: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_38) > 0
  - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
  - webnorm_invariant_gen_var_38: log_data.arguments.contactsId
- 220: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_50) > 0
  - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
  - webnorm_invariant_gen_var_50: log_data.arguments.tripId
- 231: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName

_api_v1_preserveotherservice_preserveOther@@_api_v1_consignservice_consigns_account

- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_32)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_32: log_data.arguments.foodType
- 4: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_43)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_43: log_data.env.is_user
- 101: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_29) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_29: log_data.arguments.date
- 102: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_30) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_30: log_data.arguments.foodName

_api_v1_preserveotherservice_preserveOther@@_api_v1_consignservice_consigns_order

- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_20)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_20: log_data.arguments.foodType
- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_31)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_31: log_data.env.is_user
- 52: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_17) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_17: log_data.arguments.date
- 53: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_18) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_18: log_data.arguments.foodName

_api_v1_preserveotherservice_preserveOther@@_api_v1_contactservice_contacts_account

- 1: webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_18)

- webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
- webnorm_invariant_gen_var_18: log_data.arguments.accountId

_api_v1_preserveotherservice_preserveOther@@_api_v1_foodservice_foods

- 9: webnorm_invariant_gen_var_8.equals(webnorm_invariant_gen_var_32)
  - webnorm_invariant_gen_var_8: extra@@log_data.env.user_id
  - webnorm_invariant_gen_var_32: log_data.arguments.accountId

_api_v1_preserveotherservice_preserveOther@@_api_v1_orderOtherService_orderOther_refresh

- 7: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_44: log_data.arguments.foodType
- 8: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_55)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_55: log_data.env.is_user
- 171: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_41) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_41: log_data.arguments.date
- 172: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_42) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_42: log_data.arguments.foodName

_api_v1_preserveotherservice_preserveOther@@_api_v1_orderservice_order_refresh

- 7: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_44: log_data.arguments.foodType
- 8: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_55)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_55: log_data.env.is_user
- 173: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_41) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_41: log_data.arguments.date
- 174: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_42) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_42: log_data.arguments.foodName

_api_v1_preserveotherservice_preserveOther@@_api_v1_preserveotherservice_preserveOther

- 2: webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_50)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.accountId
  - webnorm_invariant_gen_var_50: log_data.env.user_id

_api_v1_preserveotherservice_preserveOther@@_api_v1_travelservice_trips_left

- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_23: extra@@log_data.response.status
  - webnorm_invariant_gen_var_35: log_data.arguments.foodType

- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_46)
  - webnorm_invariant_gen_var_23: extra@@log_data.response.status
  - webnorm_invariant_gen_var_46: log_data.env.is_user
- 143: webnorm_invariant_gen_var_24.compareTo(webnorm_invariant_gen_var_32) < 0
  - webnorm_invariant_gen_var_24: extra@@log_data.response_time
  - webnorm_invariant_gen_var_32: log_data.arguments.date
- 144: webnorm_invariant_gen_var_24.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_24: extra@@log_data.response_time
  - webnorm_invariant_gen_var_33: log_data.arguments.foodName
- 48: webnorm_invariant_gen_var_1.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.departureTime
  - webnorm_invariant_gen_var_33: log_data.arguments.foodName
- 63: webnorm_invariant_gen_var_2.compareTo(webnorm_invariant_gen_var_31) > 0
  - webnorm_invariant_gen_var_2: extra@@log_data.arguments.endPlace
  - webnorm_invariant_gen_var_31: log_data.arguments.contactsId
- 70: webnorm_invariant_gen_var_2.compareTo(webnorm_invariant_gen_var_43) > 0
  - webnorm_invariant_gen_var_2: extra@@log_data.arguments.endPlace
  - webnorm_invariant_gen_var_43: log_data.arguments.tripId
- 84: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_43) > 0
  - webnorm_invariant_gen_var_3: extra@@log_data.arguments.startPlace
  - webnorm_invariant_gen_var_43: log_data.arguments.tripId

_api_v1_preserveotherservice_preserveOther@@_api_v1_users_login

- 4: webnorm_invariant_gen_var_11.equals(webnorm_invariant_gen_var_17)
  - webnorm_invariant_gen_var_11: extra@@log_data.response.data.userId
  - webnorm_invariant_gen_var_17: log_data.arguments.accountId

Appendix E23: Attack 23 Evaluation output

```
# Attack 23
## _api_v1_contactservice_contacts@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_40)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_40: log_data.env.is_user
## _api_v1_contactservice_contacts@@_api_v1_consignservice_consigns_account
- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_30)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_30: log_data.env.is_user
## _api_v1_contactservice_contacts@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_18)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_18: log_data.env.is_user
## _api_v1_contactservice_contacts@@_api_v1_orderOtherService_orderOther_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_42)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_42: log_data.env.is_user
## _api_v1_contactservice_contacts@@_api_v1_orderservice_order_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_42)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_42: log_data.env.is_user
## _api_v1_contactservice_contacts@@_api_v1_travelplanservice_travelPlan_minStation
- 47: webnorm_invariant_gen_var_7.compareTo(webnorm_invariant_gen_var_13) < 0
   - webnorm_invariant_gen_var_7: extra@@log_data.env.user_id
   - webnorm_invariant_gen_var_13: log_data.arguments.accountId
## _api_v1_contactservice_contacts@@_api_v1_travelservice_trips_left
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_33)
   - webnorm_invariant_gen_var_23: extra@@log_data.response.status
   - webnorm_invariant_gen_var_33: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_36)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_36: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_consignservice_consigns_account
- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_26)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_26: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_14)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_14: log_data.env.is_user
```

```
##
_api_v1_contactservice_contacts_account@@_api_v1_orderOtherService_orderOther_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_38)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_38: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_orderservice_order_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_38)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_38: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_travelservice_trips_left
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_29)
   - webnorm_invariant_gen_var_23: extra@@log_data.response.status
   - webnorm_invariant_gen_var_29: log_data.env.is_user
## _api_v1_foodservice_foods@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_39)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_39: log_data.env.is_user
- 73: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_33) < 0
   - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
   - webnorm_invariant_gen_var_33: log_data.arguments.path_param_0
- 76: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_36) < 0
   - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
   - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
- 116: webnorm_invariant_gen_var_8.compareTo(webnorm_invariant_gen_var_36) < 0
   - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
   - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
- 124: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_33) < 0
   - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
   - webnorm_invariant_gen_var_33: log_data.arguments.path_param_0
- 127: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_36) < 0
   - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
   - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
- 144: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_36) < 0
   - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
   - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
- 171: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_33) < 0
   - webnorm_invariant_gen_var_31: extra@@log_data.response_time
   - webnorm_invariant_gen_var_33: log_data.arguments.path_param_0
- 174: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_36) < 0
   - webnorm_invariant_gen_var_31: extra@@log_data.response_time
   - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
## _api_v1_foodservice_foods@@_api_v1_consignservice_consigns_account
- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_29)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
```

  - webnorm_invariant_gen_var_29: log_data.env.is_user
- 66: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_23) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_23: log_data.arguments.path_param_0
- 69: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_26) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_26: log_data.arguments.path_param_3
_api_v1_foodservice_foods@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_17)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_17: log_data.env.is_user
- 19: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_11) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_11: log_data.arguments.path_param_0
- 22: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_14) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_14: log_data.arguments.path_param_3
_api_v1_foodservice_foods@@_api_v1_orderOtherService_orderOther_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_41)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_41: log_data.env.is_user
- 91: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_35) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_35: log_data.arguments.path_param_0
- 94: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_38) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_38: log_data.arguments.path_param_3
_api_v1_foodservice_foods@@_api_v1_orderservice_order_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_41)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_41: log_data.env.is_user
- 92: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_35) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_35: log_data.arguments.path_param_0
- 95: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_38) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_38: log_data.arguments.path_param_3
_api_v1_foodservice_foods@@_api_v1_travelservice_trips_left
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_32)
  - webnorm_invariant_gen_var_23: extra@@log_data.response.status
  - webnorm_invariant_gen_var_32: log_data.env.is_user
- 85: webnorm_invariant_gen_var_24.compareTo(webnorm_invariant_gen_var_26) < 0
  - webnorm_invariant_gen_var_24: extra@@log_data.response_time
  - webnorm_invariant_gen_var_26: log_data.arguments.path_param_0

- 88: webnorm_invariant_gen_var_24.compareTo(webnorm_invariant_gen_var_29) < 0
  - webnorm_invariant_gen_var_24: extra@@log_data.response_time
  - webnorm_invariant_gen_var_29: log_data.arguments.path_param_3
- 33: webnorm_invariant_gen_var_1.compareTo(webnorm_invariant_gen_var_29) < 0
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.departureTime
  - webnorm_invariant_gen_var_29: log_data.arguments.path_param_3

_api_v1_preserveotherservice_preserveOther

- 4: webnorm_invariant_gen_var_2.equals("0") || webnorm_invariant_gen_var_2.equals("1")
  - webnorm_invariant_gen_var_2: log_data.arguments.assurance

_api_v1_preserveotherservice_preserveOther@@_api_v1_consignservice_consigns

- 13: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_42)
  - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_42: log_data.arguments.foodType
- 14: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_53)
  - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_53: log_data.env.is_user
- 273: daikon.Quant.fuzzy.gt(webnorm_invariant_gen_var_25, webnorm_invariant_gen_var_37)
  - webnorm_invariant_gen_var_25: extra@@log_data.response.data.price
  - webnorm_invariant_gen_var_37: log_data.arguments.consigneeWeight
- 274: daikon.Quant.fuzzy.gt(webnorm_invariant_gen_var_25, webnorm_invariant_gen_var_41)
  - webnorm_invariant_gen_var_25: extra@@log_data.response.data.price
  - webnorm_invariant_gen_var_41: log_data.arguments.foodPrice
- 275: daikon.Quant.fuzzy.gt(webnorm_invariant_gen_var_28, webnorm_invariant_gen_var_41)
  - webnorm_invariant_gen_var_28: extra@@log_data.response.data.weight
  - webnorm_invariant_gen_var_41: log_data.arguments.foodPrice
- 295: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_39) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_39: log_data.arguments.date
- 296: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName
- 302: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_50) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_50: log_data.arguments.tripId
- 114: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_39) < 0
  - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
  - webnorm_invariant_gen_var_39: log_data.arguments.date
- 115: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName
- 120: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_50) < 0
  - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
  - webnorm_invariant_gen_var_50: log_data.arguments.tripId
- 179: webnorm_invariant_gen_var_8.compareTo(webnorm_invariant_gen_var_40) < 0

  - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName
- 185: webnorm_invariant_gen_var_8.compareTo(webnorm_invariant_gen_var_50) < 0
  - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
  - webnorm_invariant_gen_var_50: log_data.arguments.tripId
- 197: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_39) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
  - webnorm_invariant_gen_var_39: log_data.arguments.date
- 198: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName
- 204: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_50) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
  - webnorm_invariant_gen_var_50: log_data.arguments.tripId
- 231: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName
- 237: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_50) < 0
  - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
  - webnorm_invariant_gen_var_50: log_data.arguments.tripId

_api_v1_preserveotherservice_preserveOther@@_api_v1_consignservice_consigns_account
- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_32)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_32: log_data.arguments.foodType
- 4: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_43)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_43: log_data.env.is_user
- 101: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_29) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_29: log_data.arguments.date
- 102: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_30) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_30: log_data.arguments.foodName
- 108: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_40: log_data.arguments.tripId

_api_v1_preserveotherservice_preserveOther@@_api_v1_consignservice_consigns_order
- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_20)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_20: log_data.arguments.foodType
- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_31)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_31: log_data.env.is_user

- 52: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_17) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_17: log_data.arguments.date
- 53: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_18) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_18: log_data.arguments.foodName
- 59: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_28) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_28: log_data.arguments.tripId

_api_v1_preserveotherservice_preserveOther@@_api_v1_inside_pay_service_inside_payment

- 94: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_30) < 0
  - webnorm_invariant_gen_var_11: extra@@log_data.response_time
  - webnorm_invariant_gen_var_30: log_data.arguments.tripId

_api_v1_preserveotherservice_preserveOther@@_api_v1_orderOtherService_orderOther_refresh

- 7: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_44: log_data.arguments.foodType
- 8: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_55)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_55: log_data.env.is_user
- 171: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_41) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_41: log_data.arguments.date
- 172: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_42) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_42: log_data.arguments.foodName
- 178: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_52) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_52: log_data.arguments.tripId

_api_v1_preserveotherservice_preserveOther@@_api_v1_orderservice_order_refresh

- 7: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_44: log_data.arguments.foodType
- 8: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_55)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_55: log_data.env.is_user
- 173: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_41) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_41: log_data.arguments.date
- 174: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_42) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time

  - webnorm_invariant_gen_var_42: log_data.arguments.foodName
- 180: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_52) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_52: log_data.arguments.tripId

_api_v1_preserveotherservice_preserveOther@@_api_v1_travelservice_trips_left

- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_23: extra@@log_data.response.status
  - webnorm_invariant_gen_var_35: log_data.arguments.foodType
- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_23, webnorm_invariant_gen_var_46)
  - webnorm_invariant_gen_var_23: extra@@log_data.response.status
  - webnorm_invariant_gen_var_46: log_data.env.is_user
- 143: webnorm_invariant_gen_var_24.compareTo(webnorm_invariant_gen_var_32) < 0
  - webnorm_invariant_gen_var_24: extra@@log_data.response_time
  - webnorm_invariant_gen_var_32: log_data.arguments.date
- 144: webnorm_invariant_gen_var_24.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_24: extra@@log_data.response_time
  - webnorm_invariant_gen_var_33: log_data.arguments.foodName
- 150: webnorm_invariant_gen_var_24.compareTo(webnorm_invariant_gen_var_43) < 0
  - webnorm_invariant_gen_var_24: extra@@log_data.response_time
  - webnorm_invariant_gen_var_43: log_data.arguments.tripId
- 48: webnorm_invariant_gen_var_1.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.departureTime
  - webnorm_invariant_gen_var_33: log_data.arguments.foodName
- 54: webnorm_invariant_gen_var_1.compareTo(webnorm_invariant_gen_var_43) < 0
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.departureTime
  - webnorm_invariant_gen_var_43: log_data.arguments.tripId

Appendix E24: Attack 24 Evaluation output

```
# Attack 24
## _api_v1_contactservice_contacts_account@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_36)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_36: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_consignservice_consigns_account
- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_26)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_26: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_14)
   - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_14: log_data.env.is_user
##
_api_v1_contactservice_contacts_account@@_api_v1_orderOtherService_orderOther_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_38)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_38: log_data.env.is_user
## _api_v1_contactservice_contacts_account@@_api_v1_orderservice_order_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_38)
   - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_38: log_data.env.is_user
## _api_v1_foodservice_foods@@_api_v1_consignservice_consigns
- 11: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_39)
   - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
   - webnorm_invariant_gen_var_39: log_data.env.is_user
- 73: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_33) < 0
   - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
   - webnorm_invariant_gen_var_33: log_data.arguments.path_param_0
- 76: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_36) < 0
   - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
   - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
- 116: webnorm_invariant_gen_var_8.compareTo(webnorm_invariant_gen_var_36) < 0
   - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
   - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
- 124: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_33) < 0
   - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
   - webnorm_invariant_gen_var_33: log_data.arguments.path_param_0
- 127: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_36) < 0
   - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
   - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
- 144: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_36) < 0
```

  - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
  - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
- 171: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_33) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_33: log_data.arguments.path_param_0
- 174: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_36) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_36: log_data.arguments.path_param_3
_api_v1_foodservice_foods@@_api_v1_consignservice_consigns_account
- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_29)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_29: log_data.env.is_user
- 66: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_23) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_23: log_data.arguments.path_param_0
- 69: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_26) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_26: log_data.arguments.path_param_3
_api_v1_foodservice_foods@@_api_v1_consignservice_consigns_order
- 1: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_17)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_17: log_data.env.is_user
- 19: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_11) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_11: log_data.arguments.path_param_0
- 22: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_14) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_14: log_data.arguments.path_param_3
_api_v1_foodservice_foods@@_api_v1_orderOtherService_orderOther_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_41)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_41: log_data.env.is_user
- 91: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_35) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_35: log_data.arguments.path_param_0
- 94: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_38) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_38: log_data.arguments.path_param_3
_api_v1_foodservice_foods@@_api_v1_orderservice_order_refresh
- 6: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_41)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_41: log_data.env.is_user
- 92: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_35) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time

  - webnorm_invariant_gen_var_35: log_data.arguments.path_param_0
- 95: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_38) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_38: log_data.arguments.path_param_3
_api_v1_preserveservice_preserve@@_api_v1_consignservice_consigns
- 13: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_14, webnorm_invariant_gen_var_53)
  - webnorm_invariant_gen_var_14: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_53: log_data.env.is_user
- 37: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.accountId
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 38: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.accountId
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 43: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.accountId
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 63: !webnorm_invariant_gen_var_2.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_2: extra@@log_data.arguments.consignee
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 88: !webnorm_invariant_gen_var_3.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_3: extra@@log_data.arguments.from
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 89: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_36) > 0
  - webnorm_invariant_gen_var_3: extra@@log_data.arguments.from
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 93: webnorm_invariant_gen_var_3.compareTo(webnorm_invariant_gen_var_44) > 0
  - webnorm_invariant_gen_var_3: extra@@log_data.arguments.from
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 110: !webnorm_invariant_gen_var_4.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 111: !webnorm_invariant_gen_var_4.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 113: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_39) < 0
  - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
  - webnorm_invariant_gen_var_39: log_data.arguments.date
- 114: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName
- 120: webnorm_invariant_gen_var_4.compareTo(webnorm_invariant_gen_var_50) < 0
  - webnorm_invariant_gen_var_4: extra@@log_data.arguments.handleDate
  - webnorm_invariant_gen_var_50: log_data.arguments.tripId

- 132: !webnorm_invariant_gen_var_5.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_5: extra@@log_data.arguments.id
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 133: !webnorm_invariant_gen_var_5.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_5: extra@@log_data.arguments.id
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 138: !webnorm_invariant_gen_var_5.equals(webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_5: extra@@log_data.arguments.id
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 156: !webnorm_invariant_gen_var_7.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_7: extra@@log_data.arguments.orderId
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 157: !webnorm_invariant_gen_var_7.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_7: extra@@log_data.arguments.orderId
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 162: !webnorm_invariant_gen_var_7.equals(webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_7: extra@@log_data.arguments.orderId
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 177: !webnorm_invariant_gen_var_8.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 178: !webnorm_invariant_gen_var_8.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 181: webnorm_invariant_gen_var_8.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName
- 183: !webnorm_invariant_gen_var_8.equals(webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 188: webnorm_invariant_gen_var_8.compareTo(webnorm_invariant_gen_var_50) < 0
  - webnorm_invariant_gen_var_8: extra@@log_data.arguments.phone
  - webnorm_invariant_gen_var_50: log_data.arguments.tripId
- 197: !webnorm_invariant_gen_var_9.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 198: !webnorm_invariant_gen_var_9.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 200: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_39) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
  - webnorm_invariant_gen_var_39: log_data.arguments.date
- 201: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate

  - webnorm_invariant_gen_var_40: log_data.arguments.foodName
- 203: !webnorm_invariant_gen_var_9.equals(webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 208: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_50) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.arguments.targetDate
  - webnorm_invariant_gen_var_50: log_data.arguments.tripId
- 214: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_33) > 0
  - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
  - webnorm_invariant_gen_var_33: log_data.arguments.accountId
- 216: !webnorm_invariant_gen_var_10.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 217: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_36) > 0
  - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 221: webnorm_invariant_gen_var_10.compareTo(webnorm_invariant_gen_var_44) > 0
  - webnorm_invariant_gen_var_10: extra@@log_data.arguments.to
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 231: !webnorm_invariant_gen_var_11.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 232: !webnorm_invariant_gen_var_11.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 235: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName
- 237: !webnorm_invariant_gen_var_11.equals(webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 241: webnorm_invariant_gen_var_11.compareTo(webnorm_invariant_gen_var_50) < 0
  - webnorm_invariant_gen_var_11: extra@@log_data.arguments.weight
  - webnorm_invariant_gen_var_50: log_data.arguments.tripId
- 251: !webnorm_invariant_gen_var_22.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_22: extra@@log_data.response.data.id
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 252: !webnorm_invariant_gen_var_22.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_22: extra@@log_data.response.data.id
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 257: !webnorm_invariant_gen_var_22.equals(webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_22: extra@@log_data.response.data.id
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 267: !webnorm_invariant_gen_var_23.equals(webnorm_invariant_gen_var_35)

  - webnorm_invariant_gen_var_23: extra@@log_data.response.data.orderId
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 268: !webnorm_invariant_gen_var_23.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_23: extra@@log_data.response.data.orderId
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 273: !webnorm_invariant_gen_var_23.equals(webnorm_invariant_gen_var_44)
  - webnorm_invariant_gen_var_23: extra@@log_data.response.data.orderId
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 281: daikon.Quant.fuzzy.ne(webnorm_invariant_gen_var_25, webnorm_invariant_gen_var_37)
  - webnorm_invariant_gen_var_25: extra@@log_data.response.data.price
  - webnorm_invariant_gen_var_37: log_data.arguments.consigneeWeight
- 282: daikon.Quant.fuzzy.gt(webnorm_invariant_gen_var_25, webnorm_invariant_gen_var_41)
  - webnorm_invariant_gen_var_25: extra@@log_data.response.data.price
  - webnorm_invariant_gen_var_41: log_data.arguments.foodPrice
- 284: daikon.Quant.fuzzy.gt(webnorm_invariant_gen_var_25, webnorm_invariant_gen_var_42)
  - webnorm_invariant_gen_var_25: extra@@log_data.response.data.price
  - webnorm_invariant_gen_var_42: log_data.arguments.foodType
- 288: !webnorm_invariant_gen_var_29.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_29: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 289: webnorm_invariant_gen_var_29.compareTo(webnorm_invariant_gen_var_36) > 0
  - webnorm_invariant_gen_var_29: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 294: webnorm_invariant_gen_var_29.compareTo(webnorm_invariant_gen_var_44) > 0
  - webnorm_invariant_gen_var_29: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_44: log_data.arguments.handleDate
- 302: !webnorm_invariant_gen_var_31.equals(webnorm_invariant_gen_var_35)
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_35: log_data.arguments.consigneeName
- 303: !webnorm_invariant_gen_var_31.equals(webnorm_invariant_gen_var_36)
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_36: log_data.arguments.consigneePhone
- 305: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_39) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_39: log_data.arguments.date
- 306: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_40: log_data.arguments.foodName
- 313: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_50) < 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response_time
  - webnorm_invariant_gen_var_50: log_data.arguments.tripId
_api_v1_preserveservice_preserve@@_api_v1_consignservice_consigns_account
- 3: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_43)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user

  - webnorm_invariant_gen_var_43: log_data.env.is_user
- 101: !webnorm_invariant_gen_var_21.equals(webnorm_invariant_gen_var_25)
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_25: log_data.arguments.consigneeName
- 39: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_25)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_25: log_data.arguments.consigneeName
- 40: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_26)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_26: log_data.arguments.consigneePhone
- 102: !webnorm_invariant_gen_var_21.equals(webnorm_invariant_gen_var_26)
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_26: log_data.arguments.consigneePhone
- 104: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_29) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_29: log_data.arguments.date
- 105: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_30) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_30: log_data.arguments.foodName
- 45: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_34)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_34: log_data.arguments.handleDate
- 112: webnorm_invariant_gen_var_21.compareTo(webnorm_invariant_gen_var_40) < 0
  - webnorm_invariant_gen_var_21: extra@@log_data.response_time
  - webnorm_invariant_gen_var_40: log_data.arguments.tripId
- 85: !webnorm_invariant_gen_var_19.equals(webnorm_invariant_gen_var_25)
  - webnorm_invariant_gen_var_19: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_25: log_data.arguments.consigneeName
_api_v1_preserveservice_preserve@@_api_v1_consignservice_consigns_order
- 2: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_4, webnorm_invariant_gen_var_31)
  - webnorm_invariant_gen_var_4: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_31: log_data.env.is_user
- 37: !webnorm_invariant_gen_var_5.equals(webnorm_invariant_gen_var_13)
  - webnorm_invariant_gen_var_5: extra@@log_data.env.user_id
  - webnorm_invariant_gen_var_13: log_data.arguments.consigneeName
- 38: !webnorm_invariant_gen_var_5.equals(webnorm_invariant_gen_var_14)
  - webnorm_invariant_gen_var_5: extra@@log_data.env.user_id
  - webnorm_invariant_gen_var_14: log_data.arguments.consigneePhone
- 43: !webnorm_invariant_gen_var_5.equals(webnorm_invariant_gen_var_22)
  - webnorm_invariant_gen_var_5: extra@@log_data.env.user_id
  - webnorm_invariant_gen_var_22: log_data.arguments.handleDate
- 18: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_13)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_13: log_data.arguments.consigneeName

- 19: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_14)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_14: log_data.arguments.consigneePhone
- 51: !webnorm_invariant_gen_var_9.equals(webnorm_invariant_gen_var_13)
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_13: log_data.arguments.consigneeName
- 52: !webnorm_invariant_gen_var_9.equals(webnorm_invariant_gen_var_14)
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_14: log_data.arguments.consigneePhone
- 54: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_17) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_17: log_data.arguments.date
- 55: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_18) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_18: log_data.arguments.foodName
- 24: !webnorm_invariant_gen_var_1.equals(webnorm_invariant_gen_var_22)
  - webnorm_invariant_gen_var_1: extra@@log_data.arguments.path_param_0
  - webnorm_invariant_gen_var_22: log_data.arguments.handleDate
- 62: webnorm_invariant_gen_var_9.compareTo(webnorm_invariant_gen_var_28) < 0
  - webnorm_invariant_gen_var_9: extra@@log_data.response_time
  - webnorm_invariant_gen_var_28: log_data.arguments.tripId

_api_v1_preserveservice_preserve@@_api_v1_orderOtherService_orderOther_refresh

- 128: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_42) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_42: log_data.arguments.foodName
- 67: !webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_46: log_data.arguments.handleDate
- 135: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_52) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_52: log_data.arguments.tripId
- 7: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_55)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_55: log_data.env.is_user
- 110: !webnorm_invariant_gen_var_31.equals(webnorm_invariant_gen_var_37)
  - webnorm_invariant_gen_var_31: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_37: log_data.arguments.consigneeName
- 111: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_38) > 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_38: log_data.arguments.consigneePhone
- 116: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_46) > 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_46: log_data.arguments.handleDate
- 125: !webnorm_invariant_gen_var_33.equals(webnorm_invariant_gen_var_38)

  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_38: log_data.arguments.consigneePhone
- 124: !webnorm_invariant_gen_var_33.equals(webnorm_invariant_gen_var_37)
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_37: log_data.arguments.consigneeName
- 61: !webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_37)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_37: log_data.arguments.consigneeName
- 62: !webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_38)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_38: log_data.arguments.consigneePhone
- 127: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_41) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_41: log_data.arguments.date
_api_v1_preserveservice_preserve@@_api_v1_orderservice_order_refresh
- 128: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_41) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_41: log_data.arguments.date
- 129: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_42) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_42: log_data.arguments.foodName
- 67: !webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_46)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_46: log_data.arguments.handleDate
- 126: !webnorm_invariant_gen_var_33.equals(webnorm_invariant_gen_var_38)
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_38: log_data.arguments.consigneePhone
- 7: daikon.Quant.fuzzy.eq(webnorm_invariant_gen_var_11, webnorm_invariant_gen_var_55)
  - webnorm_invariant_gen_var_11: extra@@log_data.env.is_user
  - webnorm_invariant_gen_var_55: log_data.env.is_user
- 136: webnorm_invariant_gen_var_33.compareTo(webnorm_invariant_gen_var_52) < 0
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time
  - webnorm_invariant_gen_var_52: log_data.arguments.tripId
- 111: !webnorm_invariant_gen_var_31.equals(webnorm_invariant_gen_var_37)
  - webnorm_invariant_gen_var_31: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_37: log_data.arguments.consigneeName
- 112: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_38) > 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_38: log_data.arguments.consigneePhone
- 117: webnorm_invariant_gen_var_31.compareTo(webnorm_invariant_gen_var_46) > 0
  - webnorm_invariant_gen_var_31: extra@@log_data.response.msg
  - webnorm_invariant_gen_var_46: log_data.arguments.handleDate
- 125: !webnorm_invariant_gen_var_33.equals(webnorm_invariant_gen_var_37)
  - webnorm_invariant_gen_var_33: extra@@log_data.response_time

  - webnorm_invariant_gen_var_37: log_data.arguments.consigneeName
- 61: !webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_37)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_37: log_data.arguments.consigneeName
- 62: !webnorm_invariant_gen_var_6.equals(webnorm_invariant_gen_var_38)
  - webnorm_invariant_gen_var_6: extra@@log_data.arguments.loginId
  - webnorm_invariant_gen_var_38: log_data.arguments.consigneePhone